

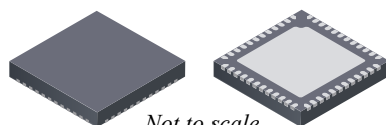
Automotive Microcontroller with LIN and MOSFET Driver

FEATURES AND BENEFITS

- 32-bit ARM cortex M4 CPU core
 - Up to 40 MHz clock frequency
 - On-chip $\pm 1\%$ accurate oscillator
 - Programmable clock generator
 - One-clock-per-machine cycle architecture
 - DMA
 - 16-level interrupt handler
 - SW-DP 2-wire debug
- On-chip memory
 - 128 kB flash
 - 64 B EEPROM
 - 32 kB DRAM
 - 8 kB IIRAM
 - 32 kB boot ROM

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PACKAGE



Not to scale

48-terminal, 7 mm $\times$ 7 mm QFN with exposed thermal pad and wettable flank (suffix EV)

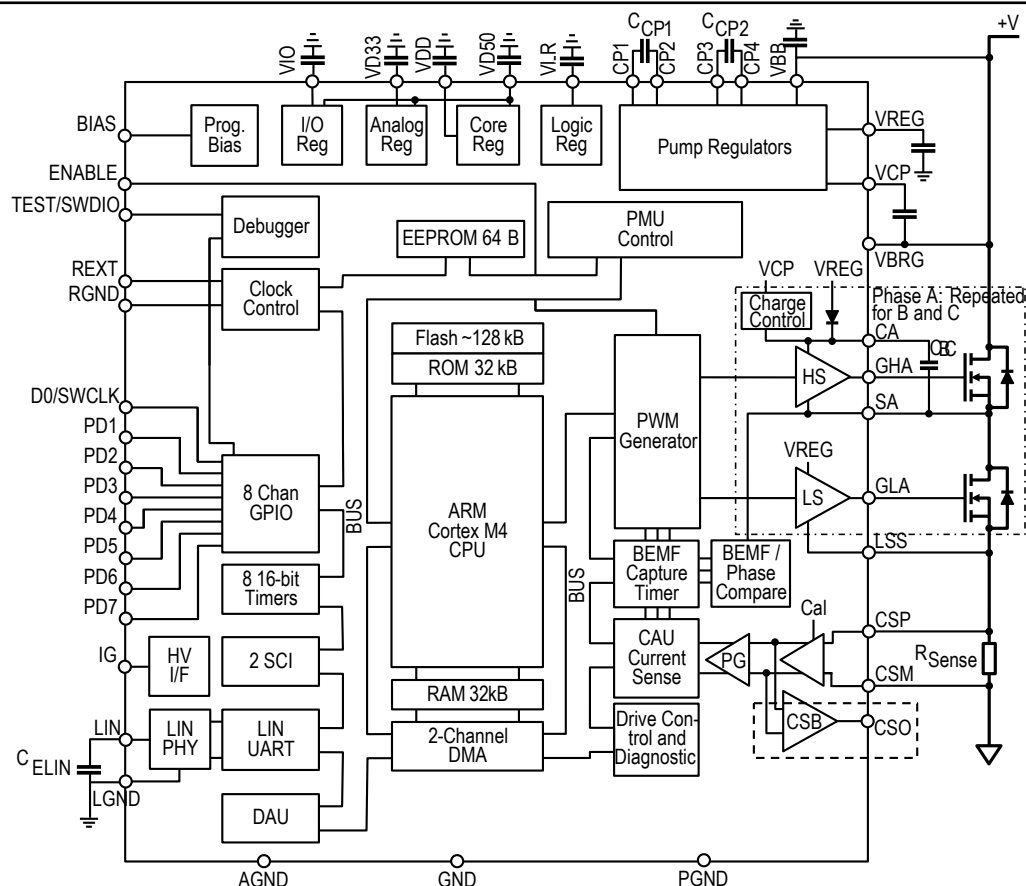
DESCRIPTION

The A89201 is a high-performance processor with LIN interface, integrated three-phase gate drive, and precision current sense capability. It is designed for use with advanced stand-alone three-phase brushless DC (BLDC) motor and permanent-magnet synchronous motor (PMSM) control applications.

The processor uses the ARM cortex M4 CPU core running at 40 MHz, giving up to 50 million instructions per second (MIPS). The processor capability is further enhanced by peripheral functions specifically designed for motor-control applications. These include a pulse-width modulation (PWM) generator and sense-current capture systems capable of providing up to 12-bit control precision at a PWM frequency of up to 20 kHz.

Eight general-purpose input and output (I/O) ports provide access to programmable serial communication interfaces and analog and digital inputs and outputs. A high-voltage input is available for supply-compliant inputs and a LIN physical interface is provided for use with LIN bus communications or for standard PWM control.

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Functional Block Diagram (see also the External Component Selection section)

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FEATURES AND BENEFITS (continued)

- Three-phase bridge MOSFET driver
 - Bootstrap gate drive for N-channel MOSFET bridge
 - Cross-conduction protection with adjustable dead time
 - Charge pump for 0% to 100% duty cycle without restriction
 - Programmable gate drive strength
- Single 5.5 – 50 V supply
- 5 V CMOS-compatible logic I/O
- 80 MHz PWM generator
 - 12-bit PWM at 20 kHz
 - Programmable back electromagnetism force (BEMF) and current-sample control
- Programmable high-performance current-sense amplifier
 - 3 x 12 bit, 1 μ s analog-to-digital converter (ADC) for current measurement
- 12-bit 1 μ s data acquisition ADC with 16-channel multiplexer (mux)
- 8 general-purpose I/O (GPIO) ports
- High-voltage ignition input (IG)
- 8 general-purpose timers
- 2 serial communication interfaces (SCI)
- 3-phase BEMF detector
- 3-phase phase-state monitor
- LIN transceiver LIN2.x/ISO17987/J2602 compatible
- LIN data-link layer
- Integrated power management
- Programmable sensor bias
- Extensive programmable diagnostics
- Automotive AEC-Q100 qualified
- Temperature Grade 1: $-40^{\circ}\text{C} \leq T_A \leq 125^{\circ}\text{C}$

DESCRIPTION (continued)

The gate driver is an N-channel power MOSFET driver capable of controlling MOSFETs connected in a 3-phase bridge arrangement and is specifically designed for automotive power applications with high-power inductive loads, such as BLDC motors.

A unique charge-pump regulator provides the supply for the MOSFET gate drive for battery voltages down to 7 V and allows the A89201 to operate with a reduced gate drive voltage down to 5.5 V. Gate drive voltage and strength are programmable to help reduce electromagnetic compatibility (EMC) issues. A bootstrap capacitor is used to provide the above-battery supply voltage required for N-channel MOSFETs. Bootstrap charge is maintained at high PWM duty cycles by an additional charge pump providing 0% to 100% PWM, without a duty-cycle restriction.

The power supply unit provides and manages all internal supplies from a single 5.5 to 50 V supply. The microcontroller unit (MCU) section can also operate with an independent single 5 V supply.

Integrated programmable diagnostics provide indication of multiple internal faults, system faults, and power bridge faults, and can be configured to protect the power MOSFETs under most short-circuit conditions.

The A89201 is supplied in a 48-pin, quad-flat no-lead (QFN) package with exposed thermal pad. This package is lead (Pb) free with 100% matte tin leadframe plating.

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SPECIFICATIONS

SELECTION GUIDE

Part Number	Flash Option (kB)	Regulator Option	Bond Option	Temperature Grade	Communication	Packing	Package
A89201KEVSR-C ^[1]	128	5 V	Resistor oscillation and CSO	Grade 1	LIN + 2 SCI (UART or SPI)	4000 pieces per 13-in. reel	7 mm × 7 mm, 0.9 mm nominal height 48-terminal QFN with exposed thermal pad and wettable flank

^[1] The following variants are also offered:

Part Number	Flash Option (kB)	Regulator Option	Bond Option	Temperature Grade	Communication
• A89201KEVSR-D	128	5 V	Resistor oscillation and FAULTn	Grade 1	LIN + 2 SCI (UART or SPI)
• A89201KEVSR-F	128	5 V	Resistor oscillation and CSO	Grade 0	LIN + 2 SCI (UART or SPI)
• A89201KEVSR-E	248	5 V	Crystal and CSO	Grade 0	CAN + 2 SCI (UART or SPI)



ABSOLUTE MAXIMUM RATINGS^{[1][2]}

Characteristic	Symbol	Conditions	Rating	Units
Supply Voltage	V _{BB}		−0.3 to 50	V
Bridge-Drain Monitor Terminal	V _{BRG}	VBRG	−0.3 to 55	V
Between-Ground Terminals		Connect xGND terminals together at package	−0.1 to 0.1	V
Drive-Regulator Terminal	V _{REG}	VREG	−0.3 to 16	V
Pump-Capacitor Terminal	V _{CP1}	CP1	−0.3 to 16	V
	V _{CP2}	CP2	−0.3 to 16	V
	V _{CP3}	CP3	−0.3 to 55.3	V
	V _{CP4}	CP4	−0.3 to 71	V
Pump-Regulator Terminal	V _{VCP}	VCP	−0.3 to 71	V
Logic-Regulator Terminal	V _{LR}	VLR	−0.3 to 6	V
Logic-Supply Terminal	VD50	VD50	−0.1 to 5.5	V
Core-Voltage Regulator Terminals	VDD	VDD	−0.1 to 1.32	V
		Respect to VD50	−5 to 0.1	V
Low-Voltage Regulator Terminal	VD33	VD33	−0.1 to 3.6	V
		Respect to VD50	−5 to 0.1	V
GPIO Voltage Regulator Terminal	V _{IO}	VIO	−0.1 to 5.5	V
		Respect to VD50	−5 to 0.1	V
Battery-Compliant Logic-Input Terminals	V _{IB}	IG	−4 to 50	V
Sensor-Bias Terminal	V _{BIAS}	BIAS	−0.3 to 50	V
GPIO Terminals	V _{PDX}	PDx	−0.1 to 5.5	V
		Respect to VD50	−5 to 0.1	V
		Respect to VIO	−5 to 0.1	V

^[1] With respect to GND. Ratings apply when another circuit operating constraint is not present.

^[2] Lowercase “x” in terminal names and symbols indicates a sequence character.

^[3] Confirmed by design and characterization.

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ABSOLUTE MAXIMUM RATINGS ^{[1][2]} (continued)

Characteristic	Symbol	Conditions	Rating	Units
Logic-Level Terminals	V_{IN}	ENABLE	-0.3 to 6	V
		TEST	-0.1 to 5.5	V
		REXT	-0.1 to 3.6	V
LIN Bus Interface	V_{LIN}	LIN	-40 to 50	V
Sense-Amplifier Input	V_{CSI}	CSP, CSM	-10 to 10	V
Sense-Amplifier Output	V_{CSO}	CSO	-0.3 to 6	V
Bootstrap-Supply Terminals	V_{CX}	CA, CB, CC	-0.3 to 71	V
High-Side Gate-Drive Output Terminals	V_{GHX}	GHA, GHB, GHC	$V_{CX} - 16$ to $V_{CX} + 0.3$	V
		GHA, GHB, GHC (transient ^[3])	-18 to $V_{CX} + 0.3$	V
Motor-Phase Terminals	V_{SX}	SA, SB, SC	$V_{CX} - 16$ to $V_{CX} + 0.3$	V
		SA, SB, SC (transient ^[3])	-18 to $V_{CX} + 0.3$	V
Low-Side Gate-drive output Terminals	V_{GLX}	GLA, GLB, GLC	$V_{REG} - 16$ to 18	V
		GLA, GLB, GLC (transient ^[3])	-8 to 18	V
Bridge Low-Side Source Terminals	V_{LSS}	LSS	$V_{REG} - 16$ to 18	V
		LSS (transient ^[3])	-8 to 18	V
GND Terminal	V_{GND}	Respect to AGND	-0.1 to 0.1	V
Transient Junction Temperature	$T_{J(trans)}$	Over temperature event not to exceed 10 seconds, lifetime duration not to exceed 10 hours, guaranteed by design characterization.	165	°C
Maximum Continuous Junction Temperature	$T_{J(max)}$		150	°C
Storage Temperature Range	T_{stg}		-55 to 150	°C

^[1] With respect to GND. Ratings apply when another circuit operating constraint is not present.

^[2] Lowercase "x" in terminal names and symbols indicates a sequence character.

^[3] Confirmed by design and characterization.

THERMAL CHARACTERISTICS ^[1]: May require derating at maximum conditions.

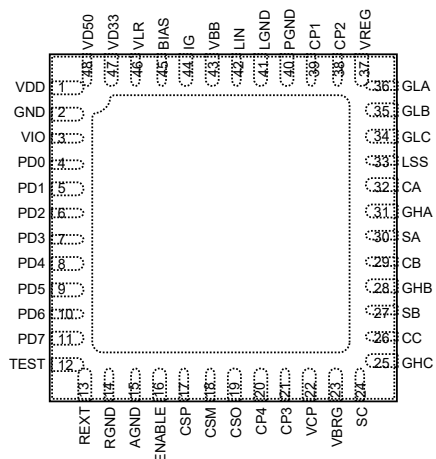
Characteristic	Symbol	Notes	Rating	Unit
Package Thermal Resistance	$R_{\theta JA}$	4-layer PCB based on JEDEC standard	23	°C/W
		2-layer PCB with 3.8 in ² copper each side	44	°C/W
	$R_{\theta JP}$		2	°C/W

^[1] Additional thermal information is available on the Allegro website.

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PINOUT DIAGRAM AND TERMINAL LIST TABLE



A89201 Package (suffix EV) Pinout

Terminal List

Terminal Name	Terminal Description	Pin Number
AGND	Quiet analog ground	15
BIAS	Sensor bias output	45
CA	Bootstrap capacitor phase A	32
CB	Bootstrap capacitor phase B	29
CC	Bootstrap capacitor phase C	26
CP1	Charge pump capacitor terminal	39
CP2	Charge pump capacitor terminal	38
CP3	Charge pump capacitor terminal	21
CP4	Charge pump capacitor terminal	20
CSM	Current-sense amplifier negative input	18
CSO	Current-sense amplifier output	19
CSP	Current-sense amplifier positive input	17
D0/SWCLK	GPIO0/SW-DP interface clock	4
PD1	GPIO1	5
PD2	GPIO2	6
PD3	GPIO3	7
PD4	GPIO4	8
PD5	GPIO5	9
PD6	GPIO6	10
PD7	GPIO7	11
ENABLE	Direct output activity control	16
GHA	High-side gate-drive phase A	31
GHB	High-side gate-drive phase B	28
GHC	High-side gate-drive phase C	25
TAB	Connect to ground	Tab

Terminal Name	Terminal Description	Pin Number
GLA	Low-side gate-drive phase A	36
GLB	Low-side gate-drive phase B	35
GLC	Low-side gate-drive phase C	34
GND	Ground	2
IG	High-voltage ignition input	44
LGND	LIN ground	41
LIN	LIN bus connection	42
LSS	Low-side source	33
PGND	Power ground	40
REXT	External resistor terminal for oscillator	13
RGND	External resistor GND terminal for oscillator	14
SA	Motor-connection phase A	30
SB	Motor-connection phase B	27
SC	Motor-connection phase C	24
TEST/SWDIO	Test/SW-DP Interface	12
VD33	Analog regulator filter capacitor	47
VBB	Main power supply	43
VBRG	High-side drain-voltage sense	23
VDD	Core-regulator filter capacitor	1
VCP	Boosted gate-drive supply	22
VD50	Processor supply	48
VIO	5 V logic I/O regulator	3
VLR	Logic regulator output	46
VREG	Gate-drive supply output	37

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External Component Selection

Capacitor	Component Type	Voltage Rating	Nominal Value	Note
C _{CP1}	Charge-pump capacitor	25 V	0.47µF	
C _{CP2}	Charge-pump capacitor	25 V	2.2 µF	
C _{VREG}	VREG decoupling capacitor [1]	25 V	check footnotes	
V _{CP}	VCP decoupling capacitor	25 V	2.2 µF	
C _{VLR}	VLR decoupling capacitor [2]	6.3 V	2.2 µF	Requires > 0.9 µF
C _{VD50}	VD50 decoupling capacitor [2]	–	check footnotes	
C _{VDD}	VDD decoupling capacitor	6.3 V	2.2 µF	Requires > 1 µF
C _{VD33}	VD33 decoupling capacitor	6.3 V	4.7 µF	Requires > 2.2 µF
C _{VIO}	VIO decoupling capacitor	6.3 V	2.2 µF	Requires > 1 µF
C _{BC}	Bootstrap capacitor [1]	25 V	check footnotes	
R _{EXT}	Oscillator reference resistor [3]	–	24.9 kΩ	Requires 10 ppm for 1% accuracy oscillation
R _{Sense}	Current sensing resistor [4]	–	check footnotes	
C _{ELIN}	LIN peripheral capacitor [5]	65 V	220 pF	
C _{VBB}	VBB decoupling capacitor [6]	65 V	100 nF	
T _{CSO}	CSO terminal	If DC current measurement is required, it must be terminated with an RC low-pass filter. Suggested values are: R = 39 kΩ C = 100 nF (5 V)		

[1] Depends on the MOSFET selection. Refer to the Selection Guide.

[2] VLR and VD50 shall be shorted for single power supply distribution; if it is shorted together, a VD50 decoupling capacitor is not required.

[3] If 1% accuracy is not required, a less-accurate resistor can be used.

[4] Depends on the peak motor-phase current requirement.

[5] According to the LIN standard.

[6] Might need to be adjusted, depending on the EMC performance requirement.

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ELECTRICAL CHARACTERISTICS: Valid at $T_J = -40$ to 150°C , $V_{BB} = 5.5$ V to 50 V (unless noted otherwise)

Characteristics	Symbol	Test Conditions	Min.	Typ.	Max.	Units
SUPPLY AND REGULATORS						
VBB Functional Operating Range	V_{BB}	Operating; outputs active	6.5	—	50	V
		Unsafe state not present	0	—	50	V
V_{BB} Quiescent Current [1]	I_{BBQ}	$V_{BB} = 12$ V, all gate-drive outputs low; does not include I_{VLR}	—	12	40	mA
VD50 Quiescent Current [1]	I_{VD50}	VD50 = 5 V	—	12	—	mA
V_{BB} Quiescent Sleep Current [1]	I_{BBS}	Sleep mode, LIN = 12 V, $V_{BB} = 12$ V, $I_{VLR} = 0$	—	—	15	μA
Bridge Operating Range [2]	V_{BRG}		5.5	13.6	50	V
VBRG Operating Current [1]	I_{BRG}	$V_{DST} = \text{default}$, $V_{BB} = 12$ V, 5.5 V $\leq V_{BRG} \leq 50$ V, LIN in recessive state	—	—	4	mA
VBRG Sleep Current [1]	I_{BRGQ}	Sleep mode, $V_{BRG} < 35$ V	—	—	10	μA
VREG Output Voltage	V_{REG}	$V_{BB} > 6.5$ V, $I_{VREG} = 0$ to 33 mA	9	—	11.7	V
		6 V $< V_{BB} \leq 6.5$ V, $I_{VREG} = 0$ to 20 mA	9	—	11.7	V
		5.5 V $< V_{BB} \leq 6$ V, $I_{VREG} = 0$ to 15 mA	9	—	11.7	V
VCP Voltage [1][3]	V_{CP}	$V_{CP} = V_{VCP} - V_{BRG}$, $I_{CP} \geq -7$ mA [1], $V_{REG} \geq 9$ V	7	—	—	V
Logic Regulator Output Voltage	V_{LR}	$I_{VLR} > -70$ mA [1], $V_{BB} > 6.5$ V	4.6	5	5.2	V
Logic Regulator Output Current Limit [1]	I_{LROC}		100	—	230	mA
Logic Supply Input Range	VD50		4.5	—	5.5	V
Core Voltage	VDD	VD50 > 4.5	1.08	1.2	1.32	V
Analog Voltage	VD33	VD50 > 4.5	3	3.3	3.6	V
System Start-up Time	t_{INIT}	$V_{BB} > 8$ V or $V_{REG} > 11$ V to gate drive enable; $C_{REG} = 10$ μF , $C_{BOOT} = 470$ nF From IG terminal high	—	—	10	ms
CORE						
System Clock Frequency	f_{OSC}		—	40	—	MHz
System Clock Accuracy	Ef_{OSC}	External resistor [4] is 24.9 k Ω nominal, 0.1%, $T_C = 10$ ppm/ $^\circ\text{C}$	—1	—	1	%
Internal Oscillator Accuracy	f_{OSC}		2	2.4	2.8	MHz

[1] For input and output current specifications, negative current is defined as coming out of (sourcing) the specified device terminal.

[2] Guaranteed by design and characterization.

[3] V_{CP} refers to the voltage on the VCP terminal with respect to (w.r.t.) the voltage on the VBB terminal. V_{VCP} refers to the voltage on the VCP terminal w.r.t. the device GND pin.

[4] The resistor oscillator circuit is defined in Figure 1.

[5] All test conditions assume that the channel is configured correctly for the required I/O function.

[6] GPIO equivalent circuit is shown in Figure 2 when DAU mode is selected. When DAU mode is not selected, C_{DAU} is disconnected.

[7] Guaranteed by design.

[8] Parameters are not guaranteed at greater than or less than the LIN 2.2 A operating limits of $V_{BB} = 7$ V to 18 V

[9] LIN parameter and timing diagram are shown in Figure 3.

[10] LIN bus load conditions (C_{BUS} ; R_{BUS}): 1 nF; 1 k Ω /6.8 nF; 660 Ω /10 nF; 500 Ω

[11] VDS offset is the difference between the programmed threshold, V_{DSTH} or V_{DSTL} , and the actual trip voltage.

[12] Data retention is calculated based on application mission profile.

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ELECTRICAL CHARACTERISTICS (continued): Valid at $T_J = -40$ to 150°C , $V_{BB} = 5.5$ V to 50 V (unless noted otherwise)

Characteristics	Symbol	Test Conditions	Min.	Typ.	Max.	Units
EEPROM						
Program Cycle	EEPROM _{PC}	−40°C < T _J < 35°C	5000	−	−	Cycles
		35°C < T _J < 85°C	500	−	−	Cycles
		85°C < T _J < 150°C	50	−	−	Cycles
Data Retention [12]	EEPROM _{DR}	−40°C < T _J < 85°C	25	−	−	Years
		85°C < T _J < 150°C	15	−	−	Years
FLASH						
Program Cycle	Flash _{PC}	−40°C < T _J < 125°C	10000	−	−	Cycles
Data Retention [12]	Flash _{DR}	−40°C < T _J < 125°C	15	−	−	Years
GPIO [5]						
Input Low Voltage	V _{IL}		−	−	0.3 V _{IO}	V
Input High Voltage	V _{IH}		0.7 V _{IO}	−	−	V
Input Hysteresis	V _{Ihys}		250	550	−	mV
Input Pull-Up Resistor	R _{DSON_PU}		25	50	80	kΩ
Input Pull-Down Resistor	R _{DSON_PD}		25	50	80	kΩ
Output Low Voltage	V _{OL}	Typical drive mode, I _{OL} = 1 mA	−	−	0.7	V
		Strong drive mode, I _{OL} = 5 mA	−	−	0.4	V
Output Leakage for Open Drain	I _{O_OD}	Output inactive, 0 V < V _O < V _{IO}	−1	−	1	μA
Output High Voltage	V _{OH}	Typical drive mode, I _{VIO} < 50 mA, I _{OL} = −1mA [1]	V _{IO} − 0.4	−	−	V
		Strong drive mode, I _{VIO} < 50 mA, I _{OL} = −5mA [1]	V _{IO} − 0.8	−	−	V
Input Leakage [1]	I _{IL}	0 V < V _O < V _{IO}	−3	−	3	μA
Input Capacitance [3]	C _{IN}	Not tested	−	−	10	pF
DAU Sampling Capacitance [2][6]	C _{DAU}		−	−	2.5	pF
DAU Series Resistance [2][6]	R _{DAU}		−	−	25	kΩ
CURRENT SENSE INPUT (COMMON TO BOTH AMPLIFIERS)						
Input Bias Current [1]	I _{BIAS}	V _{CM} in range, 6.5 V < V _{BB}	−200	−	0	μA
Input Offset Current [1]	I _{OS}	V _{ID} = 0, V _{CM} in range, 6.5 V < V _{BB}	−1.5	−	1.5	μA
Input Common-Mode Range (DC)	V _{CM}	V _{ID} = 0, 6.5 V < V _{BB}	−1.8	−	2	V

[1] For input and output current specifications, negative current is defined as coming out of (sourcing) the specified device terminal.

[2] Guaranteed by design and characterization.

[3] V_{CP} refers to the voltage on the VCP terminal with respect to (w.r.t.) the voltage on the V_{BB} terminal. V_{VCP} refers to the voltage on the VCP terminal w.r.t. the device GND pin.

[4] The resistor oscillator circuit is defined in Figure 1.

[5] All test conditions assume that the channel is configured correctly for the required I/O function.

[6] GPIO equivalent circuit is shown in Figure 2 when DAU mode is selected. When DAU mode is not selected, C_{DAU} is disconnected.

[7] Guaranteed by design.

[8] Parameters are not guaranteed at greater than or less than the LIN 2.2 A operating limits of $V_{BB} = 7$ V to 18 V

[9] LIN parameter and timing diagram are shown in Figure 3.

[10] LIN bus load conditions (C_{BUS} : R_{BUS}): 1 nF; 1 k Ω /6.8 nF; 660 Ω /10 nF; 500 Ω

[11] VDS offset is the difference between the programmed threshold, V_{DSTH} or V_{DSTL} , and the actual trip voltage.

[12] Data retention is calculated based on application mission profile.

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ELECTRICAL CHARACTERISTICS (continued): Valid at $T_J = -40$ to 150°C , $V_{BB} = 5.5$ V to 50 V (unless noted otherwise)

Characteristics	Symbol	Test Conditions	Min.	Typ.	Max.	Units
DIFFERENTIAL CURRENT SENSE AMPLIFIER						
Full-Scale Differential Output	V_{OD}		-1.2	—	1.2	V
Input Offset Voltage	V_{IOS}	Calibrated	-1	—	1	mV
Input Offset Voltage Drift	ΔV_{IOS}		-20	—	20	$\mu\text{V}/^\circ\text{C}$
Gain	A_V	Default power-up value	—	10	—	V/V
Gain Error	E_A	V_{CM} in range, SAG = 0 to 7	-1.6	—	1.6	%
Gain Drift	E_{AD}	V_{CM} in range	-70	—	70	ppm/ $^\circ\text{C}$
Small Signal -3 dB Bandwidth at Gain = 20 [7]	BW	$V_{IN} = 10$ mVpp, Gain = 20	1	—	—	MHz
Output Settling Time (to within 586 μV)	t_{SET}	$V_{ID} = -60$ mV to 60 mV square wave Gain = 20	—	—	1	μs
VBB Supply Ripple Rejection	PSRR	$V_{ID} = 0$ V, 100 kHz, Gain = 20 [7]	-0.36	—	0.36	mV/V
		$V_{ID} = 0$ V, Gain = 20	-0.18	—	0.18	mV/V
DC Common-Mode Rejection [7]	CMRR	V_{CM} step from 0 V to 200 mV, Gain = 20, 6.5 V < V_{BB}	-2.24	—	2.24	mV/V
AC Common-Mode Rejection [7]	CMRR	$V_{CM} = 200$ mVpp, 100 kHz, Gain = 20, 6.5 V < V_{BB}	-1.59	—	1.59	mV/V
		$V_{CM} = 200$ mVpp, 1 MHz, Gain = 20, 6.5 V < V_{BB}	-15	—	15	mV/V
		$V_{CM} = 200$ mVpp, 10 MHz, Gain = 20, 6.5 V < V_{BB}	-80	—	80	mV/V
Input Overload Recovery DSA	t_{IDrec}	SDG[2:0] = 2, Gain = 10, $C_{(DSAP,GND)} = 20$ pF, $C_{(DSAM,GND)} = 20$ pF, 6.5 V < V_{BB}	0.335	0.508	0.671	μs

[1] For input and output current specifications, negative current is defined as coming out of (sourcing) the specified device terminal.

[2] Guaranteed by design and characterization.

[3] V_{CP} refers to the voltage on the VCP terminal with respect to (w.r.t.) the voltage on the VBB terminal. V_{VCP} refers to the voltage on the VCP terminal w.r.t. the device GND pin.

[4] The resistor oscillator circuit is defined in Figure 1.

[5] All test conditions assume that the channel is configured correctly for the required I/O function.

[6] GPIO equivalent circuit is shown in Figure 2 when DAU mode is selected. When DAU mode is not selected, C_{DAU} is disconnected.

[7] Guaranteed by design.

[8] Parameters are not guaranteed at greater than or less than the LIN 2.2 A operating limits of $V_{BB} = 7$ V to 18 V

[9] LIN parameter and timing diagram are shown in Figure 3.

[10] LIN bus load conditions (C_{BUS} , R_{BUS}): 1 nF; 1 k Ω /6.8 nF; 660 Ω /10 nF; 500 Ω

[11] VDS offset is the difference between the programmed threshold, V_{DSTH} or V_{DSTL} , and the actual trip voltage.

[12] Data retention is calculated based on application mission profile.

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Automotive Microcontroller with LIN and MOSFET Driver

ELECTRICAL CHARACTERISTICS (continued): Valid at $T_J = -40$ to 150°C , $V_{BB} = 5.5$ V to 50 V (unless noted otherwise)

Characteristics	Symbol	Test Conditions	Min.	Typ.	Max.	Units
CSO CURRENT SENSE AMPLIFIER: GROUND-REFERENCED OUTPUT						
Input Offset Voltage	V_{IOS}		-10	—	10	mV
Input Offset Voltage Drift	ΔV_{IOS}		-40	—	40	$\mu\text{V}/^\circ\text{C}$
Gain	A_V	Default power-up value	—	20	—	V/V
Gain Error	E_A	V_{CM} in range, $6.5\text{ V} < V_{BB}$	-1.6	—	1.6	%
Gain Drift [2]	E_{AD}	V_{CM} in range, $6.5\text{ V} < V_{BB}$	-150	—	150	ppm/ $^\circ\text{C}$
Pedestal Voltage	$V_{pedestal}$	Default power-up value	—	1.2	—	V
Pedestal Voltage Error	E_{VP}	V_{CM} in range, $V_{pedestal} > 0$	-75	± 25	75	mV
Pedestal Voltage Drift	V_{PD}	V_{CM} in range, $V_{pedestal} > 0$, and $V_{pedestal} \leq 1\text{ V}$	-150	—	150	$\mu\text{V}/^\circ\text{C}$
		V_{CM} in range, $V_{pedestal} > 1\text{ V}$, and $V_{pedestal} \leq 2.4\text{ V}$	-300	—	300	$\mu\text{V}/^\circ\text{C}$
Small Signal -3dB Bandwidth at Gain = 20 [7]	BW	$V_{IN} = 10\text{ mVpp}$, $6.5\text{ V} < V_{BB}$	—	—	50	kHz
Output Dynamic Range	V_{CSOUT}	$-100\text{ }\mu\text{A} < I_{CSO} < 100\text{ }\mu\text{A}$ [1], $6.5\text{ V} < V_{BB}$	0.35	—	4.75	V
VBB Supply Ripple Rejection	PSRR	$V_{ID} = 0\text{ V}$, 100 kHz, Gain = 20, $6.5\text{ V} < V_{BB}$ [7]	-0.36	—	0.36	mV/V
		$V_{CSP} = V_{CSM} = 0\text{ V}$, DC, Gain = 20, $6.5\text{ V} < V_{BB}$	-0.18	—	0.18	mV/V
DC Common-Mode Rejection	CMRR	V_{CM} step from 0 V to 200 mV, Gain = 20, $6.5\text{ V} < V_{BB}$	-2.24	—	2.24	mV/V
AC Common-Mode Rejection [7]	CMRR	$V_{CM} = 200\text{ mVpp}$, 100 kHz, Gain = 20, $6.5\text{ V} < V_{BB}$	-1.59	—	1.59	mV/V
		$V_{CM} = 200\text{ mVpp}$, 1 MHz, Gain = 20, $6.5\text{ V} < V_{BB}$	-15	—	15	mV/V
		$V_{CM} = 200\text{ mVpp}$, 10 MHz, Gain = 20, $6.5\text{ V} < V_{BB}$	-80	—	80	mV/V

[1] For input and output current specifications, negative current is defined as coming out of (sourcing) the specified device terminal.

[2] Guaranteed by design and characterization.

[3] V_{CP} refers to the voltage on the VCP terminal with respect to (w.r.t.) the voltage on the VBB terminal. V_{VCP} refers to the voltage on the VCP terminal w.r.t. the device GND pin.

[4] The resistor oscillator circuit is defined in Figure 1.

[5] All test conditions assume that the channel is configured correctly for the required I/O function.

[6] GPIO equivalent circuit is shown in Figure 2 when DAU mode is selected. When DAU mode is not selected, C_{DAU} is disconnected.

[7] Guaranteed by design.

[8] Parameters are not guaranteed at greater than or less than the LIN 2.2 A operating limits of $V_{BB} = 7\text{ V}$ to 18 V .

[9] LIN parameter and timing diagram are shown in Figure 3.

[10] LIN bus load conditions (C_{BUS} , R_{BUS}): 1 nF; 1 k Ω /6.8 nF; 660 Ω /10 nF; 500 Ω

[11] VDS offset is the difference between the programmed threshold, V_{DSTH} or V_{DSTL} , and the actual trip voltage.

[12] Data retention is calculated based on application mission profile.

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Automotive Microcontroller with LIN and MOSFET Driver

ELECTRICAL CHARACTERISTICS (continued): Valid at $T_J = -40$ to 150°C , $V_{BB} = 5.5$ V to 50 V (unless noted otherwise)

Characteristics	Symbol	Test Conditions	Min.	Typ.	Max.	Units
CURRENT ACQUISITION ADC						
Differential Input Voltage Range ^[7]	V_{IN}	DIS = 1	-1.2	—	1.2	V
ADC Resolution ^[7]	N_{RES}	Monotonic	—	12	—	Bits
LSB ^[7]	V_{LSB}		—	586	—	μV
INL ^[7]	N_{INL}		-4	—	4	LSB
DNL ^[7]	N_{DNL}		-1	—	1	LSB
Offset Error ^[7]	V_{OFFSET}		—	1	—	LSB
Offset Thermal Drift ^[7]	V_{TEMP}		—	± 10	—	$\mu\text{V}/^\circ\text{C}$
Gain Error ^[7]	G_{ERR}		—	± 0.25	—	%
Gain Thermal Drift ^[7]	G_{DRF}		—	± 20	—	ppm/ $^\circ\text{C}$
Minimum Sample Time ^[7]	t_{SH}		—	—	1	μs
Minimum Conversion Time ^[7]	t_{CNV}		—	—	1	μs
DATA ACQUISITION ADC (DAU)						
Input Voltage Range ^[7]	V_{INDAU}		0	—	2.4	V
Input Current ^{[1][7]}	I_{IN}		—	—	1	μA
ADC Resolution ^[7]	N_{RES}	Monotonic	—	12	—	Bits
LSB ^[7]	V_{LSB}		—	586	—	μV
INL ^[7]	N_{INL}		-4	—	4	LSB
DNL ^[7]	N_{DNL}		-4	—	4	LSB
Offset Error ^[7]	V_{OFFSET}		—	1	—	LSB
Offset Thermal Drift ^[7]	V_{TEMP}		—	5	—	$\mu\text{V}/^\circ\text{C}$
Gain Error ^[7]	G_{ERR}		—	± 0.01	—	% V_{IN}
Gain Thermal Drift ^[7]	G_{DRF}		—	1	—	ppm/ $^\circ\text{C}$
Minimum Sample Time ^[7]	t_{SH}		—	—	1	μs
Conversion Time ^[7]	t_{CNV}		—	—	1	μs

^[1] For input and output current specifications, negative current is defined as coming out of (sourcing) the specified device terminal.

^[2] Guaranteed by design and characterization.

^[3] V_{CP} refers to the voltage on the VCP terminal with respect to (w.r.t.) the voltage on the VBB terminal. V_{VCP} refers to the voltage on the VCP terminal w.r.t. the device GND pin.

^[4] The resistor oscillator circuit is defined in Figure 1.

^[5] All test conditions assume that the channel is configured correctly for the required I/O function.

^[6] GPIO equivalent circuit is shown in Figure 2 when DAU mode is selected. When DAU mode is not selected, C_{DAU} is disconnected.

^[7] Guaranteed by design.

^[8] Parameters are not guaranteed at greater than or less than the LIN 2.2 A operating limits of $V_{BB} = 7$ V to 18 V

^[9] LIN parameter and timing diagram are shown in Figure 3.

^[10] LIN bus load conditions (C_{BUS} ; R_{BUS}): 1 nF; 1 k Ω /6.8 nF; 660 Ω /10 nF; 500 Ω

^[11] VDS offset is the difference between the programmed threshold, V_{DSTH} or V_{DSTL} , and the actual trip voltage.

^[12] Data retention is calculated based on application mission profile.

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Automotive Microcontroller with LIN and MOSFET Driver

ELECTRICAL CHARACTERISTICS (continued): Valid at $T_J = -40$ to 150°C , $V_{BB} = 5.5$ V to 50 V (unless noted otherwise)

Characteristics	Symbol	Test Conditions	Min.	Typ.	Max.	Units
ANALOG MULTIPLEXER INPUTS (AOUT)						
AOUT Voltage Range	V_{AOUT}		0	—	2.4	V
AOUT Settling Time	t_{AOUTs}		—	—	1	μs
Supply Voltage (VBB) Measure Range	V_{VBBM}		0	—	47.5	V
Supply Voltage (VBRG) Measure Range	V_{BRGM}		0	—	50	V
Phase Voltage (Sx) Measure Range	V_{SXM}		0	—	50	V
Regulator Voltage (VREG) Measure Range	V_{REGM}		0	—	14	V
Regulator Voltage (VCP) Measure Range	V_{CPM}		0	—	60	V
Regulator Voltage (VLR) Measure Range	V_{LRM}		0	—	5.5	V
Voltage (IG) Measure Range	V_{IG}		0	—	47.5	V
Temperature (T_J) Measure Range	T_{JM}		-50	—	190	$^\circ\text{C}$
AOUT Divider Ratio (VBB, IG)	R_{div}	$A_{OUT} = V_{XXXM} \times (R_{div} \times 100)$	—	4.76	—	%
AOUT Divider Ratio (VBRG, Sx)	R_{div}	$A_{OUT} = V_{XXXM} \times (R_{div} \times 100)$	—	4.36	—	%
AOUT Accuracy (VBB, VBRG, Sx, VCP, IG)	E_{AOUT}		-5	—	5	%
AOUT Divider Ratio (VCP)	R_{div}	$A_{OUT} = V_{XXXM} \times (R_{div} \times 100)$	—	3.46	—	%
AOUT Divider Ratio (VREG)	R_{div}	$A_{OUT} = V_{XXXM} \times (R_{div} \times 100)$	—	15	—	%
AOUT Accuracy (VREG)	E_{AOUT}		-5	—	5	%
AOUT Divider Ratio (VLR)	R_{div}	$A_{OUT} = V_{XXXM} \times (R_{div} \times 100)$	—	43.63	—	%
AOUT Accuracy (VLR)	E_{AOUT}		-2	—	2	%
AOUT Divider Ratio (T_J)	R_{div}	$A_{OUT} = V_{XXXM}$	—	1	—	V/V
AOUT Accuracy (T_J) [7]	E_{AOUT}		-10	—	10	%
IGNITION TERMINAL (IG)						
Input Low voltage	V_{IL}		—	—	2.1	V
Input High Voltage	V_{IH}		4.5	—	—	V
Input Hysteresis	V_{Ihys}		250	—	—	mV
Input Current	I_G		—	—	100	μA
Input Pull-Down Resistor	R_{PD_IG}		—	300	—	k Ω

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[2] Guaranteed by design and characterization.

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[4] The resistor oscillator circuit is defined in Figure 1.

[5] All test conditions assume that the channel is configured correctly for the required I/O function.

[6] GPIO equivalent circuit is shown in Figure 2 when DAU mode is selected. When DAU mode is not selected, C_{DAU} is disconnected.

[7] Guaranteed by design.

[8] Parameters are not guaranteed at greater than or less than the LIN 2.2 A operating limits of $V_{BB} = 7$ V to 18 V

[9] LIN parameter and timing diagram are shown in Figure 3.

[10] LIN bus load conditions (C_{BUS} ; R_{BUS}): 1 nF; 1 k Ω /6.8 nF; 660 Ω /10 nF; 500 Ω

[11] VDS offset is the difference between the programmed threshold, V_{DSTH} or V_{DSTL} , and the actual trip voltage.

[12] Data retention is calculated based on application mission profile.

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Automotive Microcontroller with LIN and MOSFET Driver

ELECTRICAL CHARACTERISTICS (continued): Valid at $T_J = -40$ to 150°C , $V_{BB} = 5.5$ V to 50 V (unless noted otherwise)

Characteristics	Symbol	Test Conditions	Min.	Typ.	Max.	Units
SENSOR BIAS TERMINAL (BIAS)						
Leakage Current	$I_{\text{leak_BIAS}}$	Bias = 50 V, $V_{BB} = 50$ V, BIAS terminal is disabled.	-1	—	1	μA
Bias Current Output, 25 μA	$I_{\text{cur_25}}$	$V_{BB} = 12$ V	22.25	25	27.75	μA
Bias Current Output, 50 μA	$I_{\text{cur_50}}$	$V_{BB} = 12$ V	43.42	50	56.58	μA
Bias Current Output, 100 μA	$I_{\text{cur_100}}$	$V_{BB} = 12$ V	87.04	100	112.96	μA
Bias Current Output, 250 μA	$I_{\text{cur_250}}$	$V_{BB} = 12$ V	209.59	250	290.41	μA
Bias Current Output, 500 μA	$I_{\text{cur_500}}$	$V_{BB} = 12$ V	417.15	500	582.85	μA
Bias Current Output, 1 mA	$I_{\text{cur_1m}}$	$V_{BB} = 12$ V	0.83	1	1.17	mA
Bias Current Output, 2.5 mA	$I_{\text{cur_2p5m}}$	$V_{BB} = 12$ V	2.1	2.5	2.9	mA
Bias Output Voltage	V_{out}	$V_{BB} = 12$ V, $I_{\text{load}} = 10$ mA	4.45	4.7	5.05	V
		$V_{BB} = 6.5$ V, $I_{\text{load}} = 10$ mA	3.3	—	4.9	V
Bias Current Limit	$I_{\text{BIAS_LIM}}$	Voltage mode	19	—	33	mA
BEMF AND PHASE COMPARATORS						
BEMF Comparator offset	V_{BO}	$5.5 \text{ V} \leq V_{BB} \leq 12 \text{ V}$	-40	—	40	mV
		$12 \text{ V} < V_{BB}$	-0.3	—	0.3	%
Phase Comparator threshold	V_{PT}		45	50	55	% V_{BRG}
Phase Comparator offset	V_{sxo}	$5.5 \text{ V} < V_{BB} < 28 \text{ V}$	-650	—	50	mV
Phase Comparator Response Time	$t_{\text{P_res}}$		—	—	200	ns

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[4] The resistor oscillator circuit is defined in Figure 1.

[5] All test conditions assume that the channel is configured correctly for the required I/O function.

[6] GPIO equivalent circuit is shown in Figure 2 when DAU mode is selected. When DAU mode is not selected, C_{DAU} is disconnected.

[7] Guaranteed by design.

[8] Parameters are not guaranteed at greater than or less than the LIN 2.2 A operating limits of $V_{BB} = 7$ V to 18 V

[9] LIN parameter and timing diagram are shown in Figure 3.

[10] LIN bus load conditions (C_{BUS} , R_{BUS}): 1 nF; 1 k Ω /6.8 nF; 660 Ω /10 nF; 500 Ω

[11] VDS offset is the difference between the programmed threshold, V_{DSTH} or V_{DSTL} , and the actual trip voltage.

[12] Data retention is calculated based on application mission profile.

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Automotive Microcontroller with LIN and MOSFET Driver

ELECTRICAL CHARACTERISTICS (continued): Valid at $T_J = -40$ to 150°C , $V_{BB} = 5.5$ V to 50 V (unless noted otherwise)

Characteristics	Symbol	Test Conditions	Min.	Typ.	Max.	Units
LIN/PWM INTERFACE BUS TRANSMITTER [8]						
Bus Recessive Output Voltage	V_{BUSRO}	LTX high, bus open load	$0.8 V_{BB}$	—	—	V
Bus Dominant Output Voltage	V_{BUSDO}	LTX low, $R_{LIN} = 500 \Omega$, $V_{BB} = 7$ V	—	—	1.4	V
		LTX low, $R_{LIN} = 500 \Omega$, $V_{BB} = 18$ V	—	—	2	V
Bus Short-Circuit Current	I_{BUSLIM}	$V_{BUS} = 18$ V	40	—	100	mA
Leakage Current, Dominant [1]	$I_{BUS_PAS_dom}$	$V_{BB} = 12$ V, $V_{BUS} = 0$ V	—1	—	—	mA
Leakage Current, Recessive	$I_{BUS_PAS_rec}$	$7 \text{ V} < V_{BB} < 18 \text{ V}$, $7 \text{ V} < V_{BUS} < 18 \text{ V}$ $V_{BUS} \geq V_{BB}$	—	—	20	μA
Leakage Current, Ground Disconnect [1]	$I_{BUS_NO_GND}$	$V_{BB} = 12$ V, $0 \text{ V} < V_{BUS} < 18 \text{ V}$ $V_{BB} = V_{GND}$, $V_{BRG} = V_{GND}$	—1	—	1	mA
Leakage Current, Supply Disconnects	$I_{BUS_NO_BAT}$	$V_{BB} = 0$ V, $0 \text{ V} < V_{BUS} < 18 \text{ V}$	—	—	100	μA
Bus Pull-Up Resistance	$R_{Peripheral}$	Typical operation	20	30	60	$\text{k}\Omega$
		Sleep state	—	2	—	$\text{M}\Omega$
LIN Input Capacitance [7]	$C_{LIN, IN}$		—	—	35	pF
LIN/PWM INTERFACE BUS RECEIVER [8][9]						
Receiver Center Voltage	V_{BUSCNT}	$V_{BUSCNT} = (TH_{Dom} + TH_{Rec})/2$	$0.475 V_{BB}$	$0.5 V_{BB}$	$0.525 V_{BB}$	V
Receiver Dominant State	V_{BUSdom}		—	—	$0.4 V_{BB}$	V
Receiver Recessive State	V_{BUSrec}		$0.6 V_{BB}$	—	—	V
Receiver Hysteresis	V_{HYS}	$V_{HYS} = TH_{Rec} - TH_{Dom}$	$0.05 V_{BB}$	—	$0.175 V_{BB}$	V
Receiver Wake-Up Threshold Voltage	V_{BUSwk}		$0.4 V_{BB}$	$0.5 V_{BB}$	$0.6 V_{BB}$	V

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[4] The resistor oscillator circuit is defined in Figure 1.

[5] All test conditions assume that the channel is configured correctly for the required I/O function.

[6] GPIO equivalent circuit is shown in Figure 2 when DAU mode is selected. When DAU mode is not selected, C_{DAU} is disconnected.

[7] Guaranteed by design.

[8] Parameters are not guaranteed at greater than or less than the LIN 2.2 A operating limits of $V_{BB} = 7$ V to 18 V

[9] LIN parameter and timing diagram are shown in Figure 3.

[10] LIN bus load conditions (C_{BUS} , R_{BUS}): 1 nF; 1 k Ω /6.8 nF; 660 Ω /10 nF; 500 Ω

[11] VDS offset is the difference between the programmed threshold, V_{DSTH} or V_{DSTL} , and the actual trip voltage.

[12] Data retention is calculated based on application mission profile.

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Automotive Microcontroller with LIN and MOSFET Driver

ELECTRICAL CHARACTERISTICS (continued): Valid at $T_J = -40$ to 150°C , $V_{BB} = 5.5$ V to 50 V (unless noted otherwise)

Characteristics	Symbol	Test Conditions	Min.	Typ.	Max.	Units
LIN/PWM INTERFACE TIMING PARAMETERS [8][9]						
Receiver Propagation Delay $H \geq L$	t_{rx_pdf}	Bus dominant to LRX low	—	—	6	μs
Receiver Propagation Delay $L \geq H$	t_{rx_pdr}	Bus dominant to LRX high	—	—	6	μs
Receiver Delay Symmetry	t_{rx_sym}	$t_{rx_pdf} - t_{rx_pdr}$	-2	—	2	μs
Bus Dominant Time for Wake	t_{BUSWK}		30	—	150	μs
Wake-Up Delay	t_{WL}	LIN wake-up to V_{REG} 90%	—	3	—	ms
Transmit Dominant Timeout	t_{TXTO}	OPM = 0	—	15	—	ms
Duty Cycle D1 (Worst Case at 20 kb/s) [10]	D1	$7\text{ V} < V_{BB} < 18\text{ V}$, $t_{BIT} = 50\text{ }\mu\text{s}$ $TH_{Rec(max)} = 0.744 V_{BB}$, $TH_{Dom(max)} = 0.581 V_{BB}$ $D1 = t_{BUS_rec(min)} / (2 \times t_{BIT})$	0.396	—	—	—
Duty Cycle D2 (Worst Case at 20 kb/s) [10]	D2	$7\text{ V} < V_{BB} < 18\text{ V}$, $t_{BIT} = 50\text{ }\mu\text{s}$ $TH_{Rec(min)} = 0.422 V_{BB}$, $TH_{Dom(min)} = 0.284 V_{BB}$ $D2 = t_{BUS_rec(max)} / (2 \times t_{BIT})$	—	—	0.581	—
Duty Cycle D3 (Worst Case at 10.4 kb/s) [10]	D3	$7\text{ V} < V_{BB} < 18\text{ V}$, $t_{BIT} = 96\text{ }\mu\text{s}$ $TH_{Rec(max)} = 0.778 V_{BB}$, $TH_{Dom(max)} = 0.616 V_{BB}$ $D3 = t_{BUS_rec(min)} / (2 \times t_{BIT})$	0.417	—	—	—
Duty Cycle D4 (Worst Case at 10.4 kb/s) [10]	D4	$7\text{ V} < V_{BB} < 18\text{ V}$, $t_{BIT} = 96\text{ }\mu\text{s}$ $TH_{Rec(min)} = 0.389 V_{BB}$, $TH_{Dom(min)} = 0.251 V_{BB}$ $D4 = t_{BUS_rec(max)} / (2 \times t_{BIT})$	—	—	0.59	—

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[4] The resistor oscillator circuit is defined in Figure 1.

[5] All test conditions assume that the channel is configured correctly for the required I/O function.

[6] GPIO equivalent circuit is shown in Figure 2 when DAU mode is selected. When DAU mode is not selected, C_{DAU} is disconnected.

[7] Guaranteed by design.

[8] Parameters are not guaranteed at greater than or less than the LIN 2.2 A operating limits of $V_{BB} = 7\text{ V}$ to 18 V .

[9] LIN parameter and timing diagram are shown in Figure 3.

[10] LIN bus load conditions (C_{BUS} , R_{BUS}): 1 nF; 1 k Ω /6.8 nF; 660 Ω /10 nF; 500 Ω

[11] VDS offset is the difference between the programmed threshold, V_{DSTH} or V_{DSTL} , and the actual trip voltage.

[12] Data retention is calculated based on application mission profile.

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Automotive Microcontroller with LIN and MOSFET Driver

ELECTRICAL CHARACTERISTICS (continued): Valid at $T_J = -40$ to 150°C , $V_{BB} = 5.5$ V to 50 V (unless noted otherwise)

Characteristics	Symbol	Test Conditions	Min.	Typ.	Max.	Units
GATE OUTPUT DRIVE						
Bootstrap Boost Voltage	V_{CP}	w.r.t V_{BRG} , $I_{CP} < 7$ mA, $V_{REG} > 9$ V	$V_{REG} - 2.6$	—	V_{REG}	V
Bootstrap Diode Forward Voltage ^[2]	V_{BOOT}	$I_D = 10$ mA	0.4	0.7	1	V
		$I_D = 100$ mA, $T_J \leq 25^\circ\text{C}$	1.5	2.2	2.8	V
		$I_D = 100$ mA, $T_J = 150^\circ\text{C}$	2.4	2.9	3.5	V
Bootstrap Diode Resistance	r_D	$r_{D(100\text{mA})} = (V_{BOOT(150\text{mA})} - V_{BOOT(50\text{mA})})/100$ mA	6	11	29	Ω
Bootstrap Diode Current Limit ^[1]	I_{DBOOT}		250	500	750	mA
High-Side Gate-Drive Static Load Resistance	R_{GSH}	$T_J < 175^\circ\text{C}$	60	—	—	k Ω
		$T_J < 25^\circ\text{C}$	45	—	—	k Ω
Turn-On Rise Time	t_{rise}	$C_{LOAD} = 15$ nF, 20% to 80%	91	—	650	ns
Turn-Off Fall Time	t_{fall}	$C_{LOAD} = 15$ nF, 80% to 20%	48	—	250	ns
Pull-Up On Resistance	$R_{DS(on)UP}$	$T_J = 25^\circ\text{C}$, $I_{GH} = -150$ mA ^[1]	5.6	—	20.4	Ω
		$T_J = 150^\circ\text{C}$, $I_{GH} = -150$ mA ^[1]	9.1	—	29.5	Ω
Pull-Up Peak Source Current ^{[1][2]}	I_{PUPK}	$V_{GS} = 0$ V	-1150	—	-276	mA
Pull-Down On Resistance	$R_{DS(on)DN}$	$T_J = 25^\circ\text{C}$, $I_{GL} = 150$ mA	1.25	—	5.7	Ω
		$T_J = 150^\circ\text{C}$, $I_{GL} = 150$ mA	2.1	—	7.7	Ω
Pull-Down Peak Sink Current ^[2]	I_{PDPK}	$V_{GS} > 8$ V, $C_{LOAD} = 15$ nF	400	800	2100	mA
Turn-On Time Set-Point Range	t_R		60	—	300	ns
Minimum Turn-On Time	t_{RM}	$t_R = 0$	—	—	70	ns
Turn-On Time Mean Step Size	t_{RS}	$t_R > 0$	14	16	18	ns
Turn-On Current (1 or 2) ^[1]	I_{HRn}	$V_{GS} = 0$ V, $I_{Rn} = 15$	-511	—	-68	mA
		Programmable set point range	-120	—	-8	mA
Turn-Off Time Set-Point Range	t_R		60	—	300	ns
Minimum Turn-Off Time	t_{RM}	$t_R = 0$	—	—	70	ns
Turn-Off Time Mean Step Size	t_{RS}	$t_R > 0$	14	16	18	ns

^[1] For input and output current specifications, negative current is defined as coming out of (sourcing) the specified device terminal.

^[2] Guaranteed by design and characterization.

^[3] V_{CP} refers to the voltage on the VCP terminal with respect to (w.r.t.) the voltage on the VBB terminal. V_{VCP} refers to the voltage on the VCP terminal w.r.t. the device GND pin.

^[4] The resistor oscillator circuit is defined in Figure 1.

^[5] All test conditions assume that the channel is configured correctly for the required I/O function.

^[6] GPIO equivalent circuit is shown in Figure 2 when DAU mode is selected. When DAU mode is not selected, C_{DAU} is disconnected.

^[7] Guaranteed by design.

^[8] Parameters are not guaranteed at greater than or less than the LIN 2.2 A operating limits of $V_{BB} = 7$ V to 18 V

^[9] LIN parameter and timing diagram are shown in Figure 3.

^[10] LIN bus load conditions (C_{BUS} ; R_{BUS}): 1 nF; 1 k Ω /6.8 nF; 660 Ω /10 nF; 500 Ω

^[11] VDS offset is the difference between the programmed threshold, V_{DSTH} or V_{DSTL} , and the actual trip voltage.

^[12] Data retention is calculated based on application mission profile.

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ELECTRICAL CHARACTERISTICS (continued): Valid at $T_J = -40$ to 150°C , $V_{BB} = 5.5$ V to 50 V (unless noted otherwise)

Characteristics	Symbol	Test Conditions	Min.	Typ.	Max.	Units
Turn-Off Current (1 or 2) [1]	I_{HF_n}	$V_{GS} = 9$ V, $I_{F_n} = 15$	84	—	148	mA
		Programmable range	8	—	120	mA
GHx Output Voltage High	V_{GHH}	Bootstrap capacitor fully charged, $-10 \mu\text{A} < I_{GH} < 10 \mu\text{A}$	$V_{CX} - 0.2$	—	—	V
GHx Output Voltage Low	V_{GHL}	$-10 \mu\text{A} < I_{GH} < 10 \mu\text{A}$	—	—	$V_{SX} + 0.2$	V
GLx Output Voltage High	V_{GLH}	$-10 \mu\text{A} < I_{GL} < 10 \mu\text{A}$	$V_{REG} - 0.2$	—	—	V
GLx Output Voltage Low	V_{GLL}	$-10 \mu\text{A} < I_{GL} < 10 \mu\text{A}$	—	—	$V_{LSS} + 0.2$	V
Gate-Source Voltage, MOSFET On	$V_{GS_{on}}$	A fault is not present	V_{ROFF}	—	V_{REG}	V
GHx Passive Pull-Down	R_{GHPD}	$V_{BB} = 0$ V, $I_{GHx} = 500 \mu\text{A}$	0.03	1	2	$M\Omega$
GHx Active Pull-Down	R_{GLPD}	$V_{BB} = 0$ V, $V_{GSHx} > 3$ V	0.7	3.5	10	Ω
GLx Passive Pull-Down	R_{GHPD}	$V_{BB} = 0$ V, $I_{GLx} = 500 \mu\text{A}$	0.03	1	2	$M\Omega$
GLx Active Pull-Down	R_{GLPD}	$V_{BB} = 0$ V, $V_{GSLx} > 3$ V	0.7	3.5	10	Ω
Turn-Off Propagation Delay	$t_{P(off)}$	PWM output change to gate output change, 50 pF load	25	—	85	ns
Turn-On Propagation Delay	$t_{P(on)}$	PWM output change to gate output change, 50 pF load	25	—	85	ns
Propagation Delay Matching (Phase-to-Phase)	Dt_{PP}	Same-state change	—	5	15	ns
Propagation Delay Matching (On-to-off)	Dt_{OO}	Single phase	—	15	30	ns
Propagation Delay Matching (GHx-to-GLx)	Dt_{HL}	Same-state change	—	—	35	ns

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[4] The resistor oscillator circuit is defined in Figure 1.

[5] All test conditions assume that the channel is configured correctly for the required I/O function.

[6] GPIO equivalent circuit is shown in Figure 2 when DAU mode is selected. When DAU mode is not selected, C_{DAU} is disconnected.

[7] Guaranteed by design.

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[10] LIN bus load conditions (C_{BUS} , R_{BUS}): 1 nF; 1 k Ω /6.8 nF; 660 Ω /10 nF; 500 Ω

[11] VDS offset is the difference between the programmed threshold, V_{DSTH} or V_{DSTL} , and the actual trip voltage.

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ELECTRICAL CHARACTERISTICS (continued): Valid at $T_J = -40$ to 150°C , $V_{BB} = 5.5$ V to 50 V (unless noted otherwise)

Characteristics	Symbol	Test Conditions	Min.	Typ.	Max.	Units
DIAGNOSTICS AND PROTECTION						
VBB Undervoltage Lockout	V_{BBUV_fal}	V_{BB} falling	5.24	—	5.8	V
	V_{BBUV_ris}	V_{BB} rising	5.75	—	6.3	V
VBB POR Voltage	V_{BBR}	V_{BB} falling	3.42	3.58	3.73	V
VBB POR Hysteresis	V_{BBR_Hys}		440	550	700	mV
VCP Undervoltage	V_{CPUV}	V_{CP} falling	5	5.7	6.3	V
VCP Undervoltage Hysteresis	$V_{CPUVHys}$		100	200	400	mV
VLR Undervoltage Detection	V_{LRUV_ris}	V_{LR} rising	4.3	4.5	4.7	V
	V_{LRUV_fal}	V_{LR} falling	3.7	3.9	4.1	V
VLR Shut-Down Voltage	V_{LROSD}	V_{LR} rising	1.35	—	1.55	V
	V_{LROE}	V_{LR} falling	1.2	—	—	V
VD50 Undervoltage	$V_{VD50_UV_fal}$	VD50 undervoltage indication	3.8	4	4.2	V
	$V_{VD50_UV_ris}$	VD50 undervoltage recovery	4	4.1	4.4	V
VD33 Undervoltage	$V_{VD33_UV_fal}$	VD33 rising	2.45	2.6	2.75	V
	$V_{VD33_UV_ris}$	VD33 falling	2.6	2.7	2.9	V
VIO Undervoltage	$V_{IO_UV_5V_fal}$	V_{IO} falling	3.85	4.1	4.3	V
	$V_{IO_UV_5V_ris}$	V_{IO} rising	4.05	4.3	4.5	V
VIO Undervoltage Hysteresis	V_{IO_Hys}		0.05	—	—	V
VREG Undervoltage	V_{RON}	V_{REG} rising	7.6	7.9	8.2	V
	V_{ROFF}	V_{REG} falling	6.9	7.2	7.4	V
Bootstrap Undervoltage	V_{BCUV}	V_{BOOTx} falling, $V_{BOOTx} = V_{Cx} - V_{Sx}$	5.2	5.8	6.7	V
Bootstrap Undervoltage Hysteresis	$V_{BCUVHys}$		820	950	1100	mV
VBRG Input Voltage	V_{BRG}	When VDS monitor is active	5.5	—	50	V

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[4] The resistor oscillator circuit is defined in Figure 1.

[5] All test conditions assume that the channel is configured correctly for the required I/O function.

[6] GPIO equivalent circuit is shown in Figure 2 when DAU mode is selected. When DAU mode is not selected, C_{DAU} is disconnected.

[7] Guaranteed by design.

[8] Parameters are not guaranteed at greater than or less than the LIN 2.2 A operating limits of $V_{BB} = 7$ V to 18 V

[9] LIN parameter and timing diagram are shown in Figure 3.

[10] LIN bus load conditions (C_{BUS} ; R_{BUS}): 1 nF; 1 k Ω /6.8 nF; 660 Ω /10 nF; 500 Ω

[11] VDS offset is the difference between the programmed threshold, V_{DSTH} or V_{DSTL} , and the actual trip voltage.

[12] Data retention is calculated based on application mission profile.

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ELECTRICAL CHARACTERISTICS (continued): Valid at $T_J = -40$ to 150°C , $V_{BB} = 5.5$ V to 50 V (unless noted otherwise)

Characteristics	Symbol	Test Conditions	Min.	Typ.	Max.	Units
VDS Threshold—High Side	V_{DSTH}	Default power-up value	—	1.2	—	V
		Programmable range, $5.5 \text{ V} \leq V_{BRG} \leq 7 \text{ V}$	0	—	2.5	V
		Programmable range, $V_{BRG} > 7 \text{ V}$	0	—	3.15	V
High-Side VDS Threshold Offset [11]	V_{DSTHO}	High side on, $V_{DSTH} \geq 1 \text{ V}$, $V_{BRG} > 7 \text{ V}$	–200	—	200	mV
		High side on, $V_{DSTH} < 1 \text{ V}$	–150	—	150	mV
VDS Threshold—Low Side	V_{DSTL}	Default power-up value	—	1.2	—	V
		Programmable range	0	—	3.15	V
Low-Side VDS Threshold Offset [11]	V_{DSTLO}	Low side on, $V_{DSTL} \geq 1 \text{ V}$, $V_{BRG} > 7 \text{ V}$	–200	—	200	mV
		Low side on, $V_{DSTL} < 1 \text{ V}$	–150	—	150	mV
VDS Qualify Time	t_{VDQ}	Default power-up value	1.36	1.6	1.76	μs
		Programmable range	0.6	—	7	μs
BIAS Undervoltage Threshold	$V_{BO_UV_fal}$	Falling	3.32	3.42	3.52	V
	$V_{BO_UV_ris}$	Rising	3.6	3.75	3.9	V
Temperature Warning Threshold	T_{JWH}	Temperature increasing	125	—	160	$^\circ\text{C}$
Temperature Warning Hysteresis	T_{JWHys}		13	—	21	$^\circ\text{C}$
Overtemperature Threshold	T_{JF}	Temperature increasing	165	175	195	$^\circ\text{C}$
Overtemperature Hysteresis	T_{JHyst}	Recovery = $T_{JF} - T_{JHyst}$	11	—	20	$^\circ\text{C}$

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[2] Guaranteed by design and characterization.

[3] V_{CP} refers to the voltage on the VCP terminal with respect to (w.r.t.) the voltage on the VBB terminal. V_{VCP} refers to the voltage on the VCP terminal w.r.t. the device GND pin.

[4] The resistor oscillator circuit is defined in Figure 1.

[5] All test conditions assume that the channel is configured correctly for the required I/O function.

[6] GPIO equivalent circuit is shown in Figure 2 when DAU mode is selected. When DAU mode is not selected, C_{DAU} is disconnected.

[7] Guaranteed by design.

[8] Parameters are not guaranteed at greater than or less than the LIN 2.2 A operating limits of $V_{BB} = 7 \text{ V}$ to 18 V

[9] LIN parameter and timing diagram are shown in Figure 3.

[10] LIN bus load conditions (C_{BUS} ; R_{BUS}): 1 nF; 1 k Ω /6.8 nF; 660 Ω /10 nF; 500 Ω

[11] VDS offset is the difference between the programmed threshold, V_{DSTH} or V_{DSTL} , and the actual trip voltage.

[12] Data retention is calculated based on application mission profile.

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CHARACTERISTIC PERFORMANCE

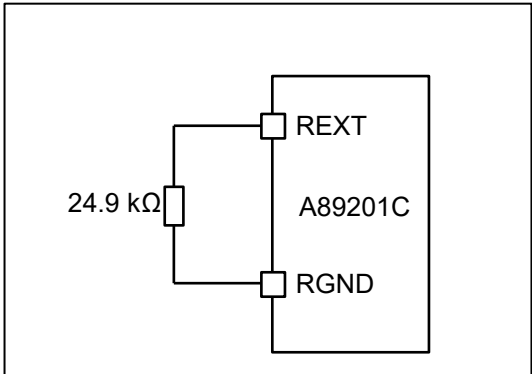


Figure 1: Resistor Oscillator External Circuit

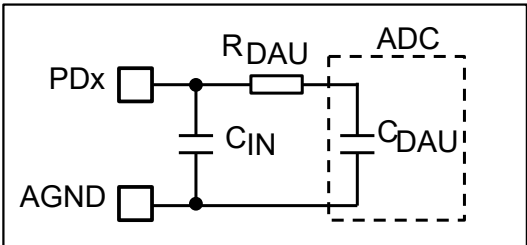


Figure 2: DAU Equivalent Circuit in ADC Mode

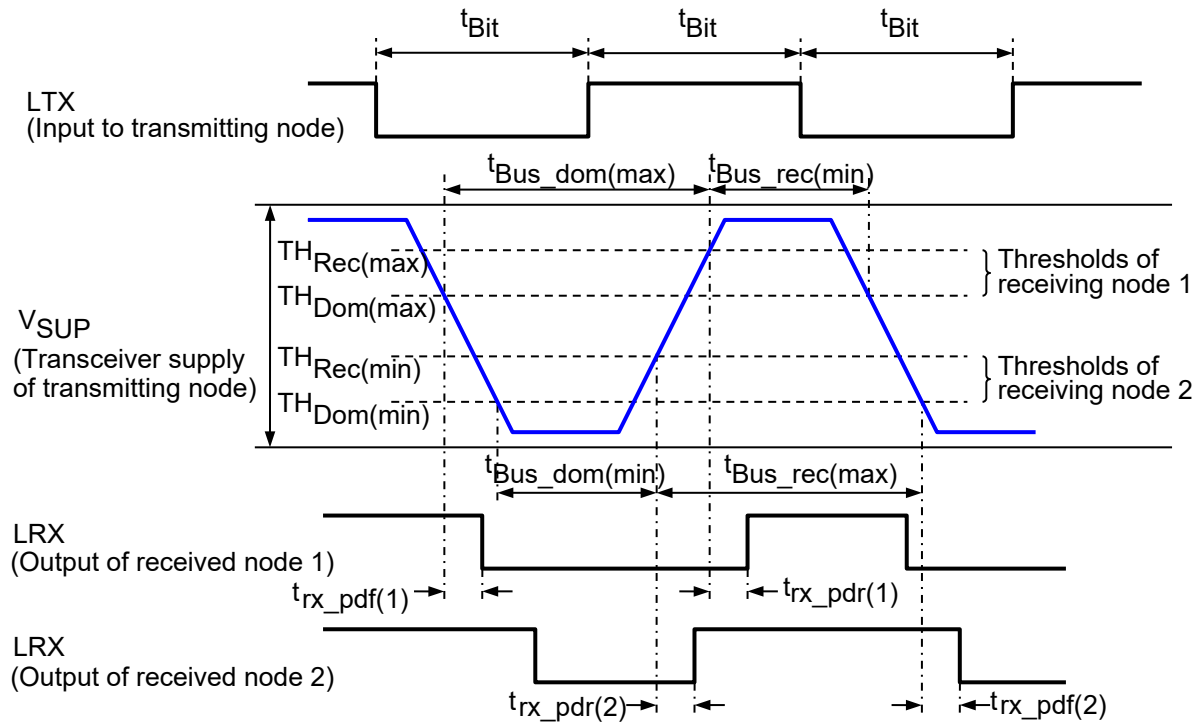


Figure 3: LIN Timing Diagram

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FUNCTIONAL DESCRIPTION

Input and Output Terminal Functions

VBB: Main power supply for the internal regulators. The main power supply should be connected to VBB through a reverse-voltage protection circuit and should be decoupled with ceramic capacitors connected close to the supply and ground terminals.

VBRG: Sense input to the top of the external MOSFET bridge. Allows accurate measurement of the voltage at the drain of the high-side MOSFETs in the bridge. Also used as the reference voltage for the bootstrap charge pump.

VREG: Regulated voltage, nominally 12 V, used to supply the low-side gate drivers and to charge the bootstrap capacitors. A sufficiently large storage capacitor must be connected to this terminal to provide the transient charging current. Refer to the VREG Capacitor Selection section.

CP1, CP2: Pump capacitor connection for bootstrap charge pump. Connect a capacitor with a recommended nominal value of 0.47 μF between CP1 and CP2. This device should have a rated working voltage of at least 25 V and a tolerance of $\pm 20\%$ or better.

VCP: Pumped gate-drive voltage referenced to the VBRG terminal. Can be used to bias external circuitry. Connect a ceramic capacitor, CVCP, of nominal value 2.2 μF between VCP and VBB. This device should have a rated working voltage of at least 25 V and a tolerance of $\pm 20\%$ or better.

CP3, CP4: Pump capacitor connection for VCP charge pump. Connect a capacitor with a recommended nominal value of 2.2 μF between CP3 and CP4. This device should have a rated working voltage of at least 25 V and a tolerance of $\pm 20\%$ or better.

VLR: Logic low-voltage dropout (LDO) regulator output. To ensure stability, a ceramic capacitor of at least 2.2 μF with an ESR of no more than 250 m Ω should be fitted and connected locally between the VLR terminal and GND.

VD50: Supply to processor. Typically connected to VLR. Connect local decoupling capacitor(s) to stabilize.

VD33: External capacitor connection for analog regulator. Connect a capacitor with a recommended nominal value of 4.7 μF between VD33 and GND. This device should have a rated working voltage of at least 6.3 V and a tolerance of $\pm 20\%$ or better.

VIO: VIO and VD50 terminals are required to be shorted and the GPIO rail operates at 5 V.

VDD: External capacitor connection for core regulator. Connect a capacitor with a recommended nominal value of 2.2 μF between VDD and GND. This device should have a rated working voltage of at least 5 V and a tolerance of $\pm 20\%$ or better.

AGND: Low-noise analog ground terminal. This terminal should not have a common impedance from the other ground terminals.

LGND: LIN bus ground terminal.

RGND: External resistor ground terminal. This terminal should not have a common impedance from the other ground terminals.

GND: Main processor ground terminal.

PGND: Power-switching ground terminal.

NOTE: Connect all ground terminals together at the A89201.

CA, CB, CC: High-side connections for the bootstrap capacitors and positive supply for high-side gate drivers. For more information, refer to the Bootstrap Capacitor Selection section.

GHA, GHB, GHC: High-side, gate-drive outputs for external N-channel MOSFETs.

SA, SB, SC: Load phase connections. These terminals sense the voltages switched across the load. They are also connected to the negative side of the bootstrap capacitors and are the negative supply connections for the floating high-side drivers.

GLA, GLB, GLC: Low-side, gate-drive outputs for external N-channel MOSFETs.

LSS: Low-side return path for discharge of the capacitance on the MOSFET gates, connected to the common source connection of the low-side external MOSFETs independently through a low-impedance track.

CSP, CSM: Current-sense amplifier inputs.

CSO: Current-sense amplifier output. This terminal is only available for option A and C.

CMPA, CMPB, CMPC: BEMF zero-crossing and bridge phase comparator digital outputs

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BIAS: Programmable sensor bias output. When the voltage mode is selected, 2.2 μF is required to stabilize the internal regulator. When the current mode is selected, 100 nF (maximum) is required to stabilize the system.

TEST/SWDIO: Mode selection input. The power-up voltage level determines the mode. In typical operation mode, the terminal should be connected to GND. When SWD is selected, the terminal should connect to VD33 so that the TEST terminal behaves as the SWDIO terminal after power-on. The terminal configuration is decided within 3 ms after power on. A power cycle is necessary to reconfigure the terminal.

REXT: Oscillator resistor. Connect the resistor between REXT and CGND. For best performance, the resistor should not include any common impedances.

PD0/SWCLK: GPIO0 or ARM SW debug clock input selectable at power on via TEST terminal.

PD1 through PD7: Standard configurable GPIO terminals. These can be configured as digital inputs and outputs, as well as DAU inputs. For more detail, refer to the GPIO section.

LIN: LIN bus connection compliant with ISO17987 (LIN 2.x). This input can also be used as a PWM control input.

IG: High-voltage input compliant with the maximum supply voltage for interface to battery-compliant vehicle controls and switches.

ENABLE: Disables all gate-drive outputs when pulled low. Provides an independent output disable, directly to the gate-drive outputs, to allow a fast disconnect of the power bridge.

Power Supplies and Regulators

The A89201 operates from a single power supply. It can also be used with a separate logic supply to the MCU section. The main power supply should be connected to VBB through a reverse-voltage protection circuit. A 100 nF ceramic decoupling capacitor must be connected from VBB close to the supply and ground terminals of the A89201.

INTERNAL REGULATORS

Several regulator circuits are included: two charge pump regulators and three linear regulators. The main charge pump regulator provides the gate-drive supply, V_{REG} , and the supply to the second pump regulator. To provide the high-side auxiliary charge voltage, the second pump regulator produces a voltage, V_{CP} , that is greater than the bridge supply voltage. The main LDO linear regulator provides the main logic supply, V_{LR} , directly from the main supply voltage, V_{BB} . This logic supply can be used to provide the logic supply voltage, V_{D50} , to the VD50 input. The supply on VD50 is fed to three LDO linear regulators to supply the core and associated logic circuits, VDD, the analog circuits, and VD33.

The output voltage of the main supply linear regulator, V_{LR} , available at the VLR terminal is 5 V. The VLR terminal is typically connected directly, or through an EMC filter, to the VD50 terminal to provide the supply for the other linear regulators. A capacitor of at least 1 μF must be connected between VLR and GND. The VD50 input can also be supplied by an external 5 V source.

The core regulator, VDD, supplied from the VD50 terminal, requires a 2.2 μF capacitor connected between VDD and GND. The analog regulator, VD33, supplied from the VD50 terminal requires a 4.7 μF capacitor connected between VD33 and GND,

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For 5 V logic, the VIO terminal needs to be shorted to the VD50 terminal.

The core logic is guaranteed to operate correctly to less than the VBB power-on reset (POR) level: When a power-on-reset state is present, this ensures that the A89201 continues to operate safely until all logic is reset.

The sense-amplifier output is supplied by and referenced to the voltage at the VLR terminal, V_{LR} .

Low Supply-Voltage Operation

The A89201 operates within specified parameters at a V_{BB} of 7 V to 50 V. At less than 6.5 V, the gate-drive outputs and the I/O might be inactive; however, the A89201 continues to function with a supply down to 3.5 V. For all supply switching conditions, it remains in a safe state between 0 V and 50 V. This provides a very rugged solution for use in the harsh automotive environment.

VREG Charge-Pump Regulator

The gate drivers are powered by an internal voltage regulator that generates a voltage, V_{REG} , at the VREG terminal. It limits the supply voltage to the drivers and, therefore, the maximum gate voltage. At low supply voltage, the regulated supply is maintained by a charge-pump boost converter.

This regulator uses a charge pump with a switching frequency of 62.5 kHz. It operates as a regulated doubler or a step-down regulator depending on the input voltage on the VBB terminal. Regulator operation requires two external capacitors: one pump capacitor and one storage capacitor. The pump capacitor, CP1, should have a nominal value of 0.47 μ F and should be connected between the CP1 and CP2 terminals. This capacitor should be of a high-quality dielectric material, such as X7R, and should have a rated working voltage of at least 25 V and a tolerance of $\pm 20\%$ or better. The storage capacitor, C_{VREG} , is connected between the VREG and GND terminals. The storage capacitor value depends on the size of the bootstrap capacitors and the total gate charge of the external power MOSFETs. Further details about selection of the value of C_{VREG} follow.

VCP Charge-Pump Regulator

This regulator generates a voltage that exceeds the bridge supply connected to the VBRG terminal. It provides the additional charge current for the bootstrap capacitors on each phase when that phase is switched at high PWM duty cycles and provides the bias current to allow the high-side MOSFET to be held in the on state.

This regulator also uses a charge-pump scheme with a switching frequency of 62.5 kHz. The voltage at the VREG terminal is pumped to produce the required output. Regulator operation requires two external capacitors: one pump capacitor and one storage capacitor. The pump capacitor, CP2, should have a nominal value of 2.2 μ F and should be connected between the CP3 and CP4 terminals. The storage capacitor, CVCP, should have a nominal value of 2.2 μ F and should be connected between the VCP and VBB terminals. Both the pump and storage capacitors should be of a high-quality dielectric material, such as X7R, and should have a rated working voltage of at least 25 V and a tolerance of $\pm 20\%$ or better.

ARM Cortex-M4 Core

At the heart of the A89201 is a Cortex-M4 CPU core, using the ARMv7 Thumb instruction set, driving a dual advanced high-performance bus (AHB) architecture. The M4 includes a rich DSP instruction set to enable processing of DSP commands within one machine cycle. This enables cycle-efficient processing of complex motor-control algorithms. A low-latency interrupt service provides support for real-time interrupt-based applications.

CPU FEATURE SET

- ARM Cortex M4 processor
- ARMv7-M Thumb instruction set
- 32-bit Harvard architecture
- DSP command set
- 32-bit hardware divider
- Dual AMBA AHB-Lite bus for ICode and DCode
- Nested vectored interrupt controller (NVIC)
 - 16 peripheral interruption
 - Configurable priority
 - Grouping priority
 - Low-latency interrupt service
- Native SW-DP 2 terminal debugger support

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BUS ARCHITECTURE

Two independent buses are provided. Each implements the AMBA AHB-Lite including the advance peripheral bus (APB) bridge. Each bus supports the Harvard architecture used by the Cortex M4 core. The architecture uses one address bus and two data buses: one for instruction (ICode) and one for data (DCode). This allows independent access for instruction fetch and data access. The bus address map is shown in Figure 4.

System Control	0xFFFF_FFFF
	0xE000_0000
Peripherals	0x4000_7FFF
	0x4000_0000
DRAM	0x2000_7FFF
	0x2000_0000
Flash Memory	0x0801_FFFF
	0x0800_0000
IRAM	0x0001_1FFF
	0x0001_0000
Boot ROM	0x0000_7FFF
	0x0000_0000

Figure 4: Bus Address Map

DMA

DMA provides the ability to use either of the buses independently from the core. Transfers are supported from memory to memory, memory to peripheral, and peripheral to memory. Transfers can be initiated by several units, including data acquisition, current acquisition, communications, and timers.

Memory

BOOT ROM

The 32 kB boot ROM in the A89201 provides the initialization code to the system. Following a power-on reset, the processor starts working from address 0x0 located in the boot ROM.

The boot code includes the following functions:

- System verification
- Peripheral initialization
- Flash program initialization
- Boot loader for SRAM

On completion of boot ROM initialization, the processor immediately starts to execute user software, starting from 0x0800_0000.

RAM

A total of 32 kB of high-speed DRAM is available starting at address 0x2000_0000 and 8 kB of IRAM is available starting at address 0x0001_0000. This can be accessed by either the core or the direct memory access (DMA).

FLASH

Up to 128 kB of Flash memory is available for program and data storage starting at address 0x0800_0000. Several methods are available for loading program code into flash memory via different routes, in addition to the standard SW-DP access.

EEPROM

An independent EEPROM memory is provided to store critical system and application configuration data on the gate-drive unit (GDU). The EEPROM can store 32 words of 16 bits (64 B). The EEPROM includes error correction and detection that can correct a single error and can detect 2-bit errors to prevent the need to transfer incorrect configuration data to the buffer.

The A89201 also can update the EEPROM contents from the buffer and rewrite the EEPROM content into the buffer. The EEPROM programming and read-transfer sequences are fully integrated into the EEPROM block and are controlled through the EEPROM control register.

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INTERRUPT CONTROL UNIT

A low-latency interrupt service controller accepts inputs from multiple sources, prioritizes these inputs, interrupts the CPU, and provides the required interrupt vector.

The A89201 provides 20 interrupt vectors with 16 priority levels. In addition to the system control interrupt sources, interrupts can be generated by any of the following internal peripherals:

- Current acquisition unit (CAU)
- Data acquisition unit (DAU)
- General timer unit (GTU)
- Serial communication interface (SCI)
- LIN communication interface (LIN)
- Gate driver unit (GDU)
- General-purpose input/output (GPIO)
- Advanced motor control timer (AMCT)
- Direct memory access (DMA)
- System control unit (SCU)

The priority of each interrupt is programmable, and each interrupt can be masked (ignored) via software control.

NONMASKABLE INTERRUPT

In addition to the maskable interrupts, there is a reserved vector for a nonmaskable interrupt (NMI). The NMI is triggered by any of the following system events:

- Flash memory double-bit error
- SRAM double-bit error
- BootROM double-bit error
- MCU over temperature
- Watchdog error

For each interrupt, several events are possible to be the cause.

CLOCK TREE

A high-precision 80 MHz clock control unit (CCU) provides the clock for the CPU and peripherals. The CCU provides the following clock sources:

- f_{MCU} : Cortex-M4 (Maximum: 40 MHz)
- f_{PWM} : PWM generator clock (Maximum: 80 MHz)
- f_{CAU} : CAU clock
- f_{DAU} : DAU clock
- f_{TIM1} : Timer 1 clock
- f_{TIM2} : Timer 2 clock
- f_{LIN} : LIN transceiver clock
- f_{SCI1} : SCI1 clock
- f_{SCI2} : SCI2 clock
- $f_{systick}$: System tick timer

RESISTOR OSCILLATOR

1% accurate oscillator is available by use of external resistor.

For robust oscillator operation, the PCB design guidelines that follow are required.

- Differential connection: From REXT to the RGND terminal, a differential connection is strongly recommended. The external resistor should connect to the device as close as practical.

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Peripheral Units

GENERAL TIMER UNIT (GTU)

The GTU provides eight general-purpose timers with selectable signal sources to enable time-based measurements of multiple input signals.

The following features are provided:

- Eight independent timers
- 16-bit counter
- Independent scalable counter clock
- Configurable interrupt
- Four measurement modes
 - Period
 - Signal high time
 - Signal low time
 - Event count
- Selectable trigger edge
- Overflow detection
- Trigger output
- PWM mode

MEASUREMENT INPUT AND COUNTER CLOCK

For each timer, the signal to be measured, the monitor signal, can be selected from any of the GPIO inputs: either SCI receive signal or the PWM center signal. Also, for each timer, the counter clock source can be selected as the timer clock from the clock control unit or from the trigger output from any timer. Each timer can also be independently disabled if required.

COUNTER MEASUREMENTS

Period measurements are made between successive rising or falling edges of the monitor signal where the edge is selectable.

Signal high measurements are made between a rising edge and the following falling edge of the monitor signal. Signal low measurements are made between a falling edge and the following rising edge of the monitor signal.

The edges of the monitor signal that are used by the event counter are selectable as rising, falling, or both. The monitor signal is sampled by the selectable clock input and each edge selected advances the count.

GENERAL-PURPOSE INPUT/OUTPUT (GPIO)

The GPIO unit provides access to eight external terminals that can be configured in multiple combinations of digital or analog inputs and digital outputs.

- Configurable signal direction
- Configurable output driving capability
- Selectable pull-up or pull-down
- CMOS-compliant input
- Open-drain support
- Configurable as DAU input
- Short-circuit protection

A functional block diagram is shown in Figure 5.

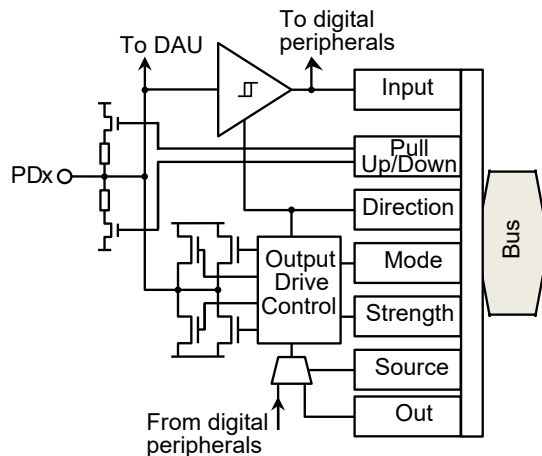


Figure 5: GPIO Block Diagram

SIGNAL DIRECTION CONFIGURATION

Each channel can be configured as input-only, output with input, or fully disabled. When configured as an input, the output becomes fully disabled and the input Schmitt buffer passes the logic input at the GPIO terminal to the bus register for access by the CPU. An optional input pull-up resistor to VD50 or pull-down resistor to GND may be enabled, if required.

When configured as an output, the input remains active to provide positive confirmation of the output state. The output state can be taken from the bus register content as entered by the CPU or by an input from one of the digital peripheral units as selected by the CPU. The output mode and drive strength can also be configured by the CPU. The output can be configured as a non-overlap push-pull drive or as an open-drain pull-down or pull-up. The drive strength can be selected as a typical, 1 mA drive, or a boost, 5 mA drive, for higher-frequency output signals.

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INTERRUPT GENERATION

The GPIO unit can be configured to provide an interrupt on each of the eight inputs. For each input, the interrupt trigger can be independently disabled or generated by a rising edge, falling edge, or both, or by a high level and a low level.

SERIAL COMMUNICATION INTERFACE (SCI)

The two general-purpose serial communication interface (SCI) units support UART, SPI controller, and SPI peripheral operation. The block diagram is shown in Figure 6.

The SCI unit supports the following functions:

- Transmit/Receive buffer from 2 to 16
- Clock polarity
- Interrupt:
 - Transmit complete
 - Receive complete
 - SCI error
- SPI primary and secondary support
- Most-significant bit (MSB) or least-significant bit (LSB) first
- Full- and half-duplex support
- Fast transfer
 - Controller up to 20 Mbps
 - Peripheral up to 2.5 Mbps
 - SPI up to 20 Mbps

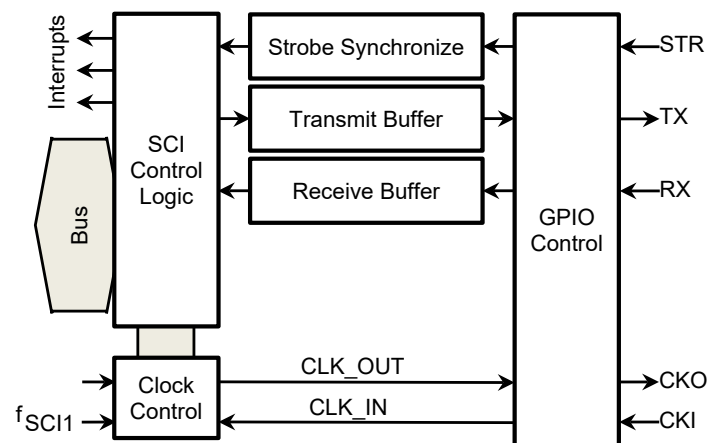


Figure 6: SCI Block Diagram

SCI REGISTERS

Mode Select: Selects the serial communication protocol.

Transmit/Receive Buffers: These are the 16-bit buffers that hold the data that is to be serialized and sent, and the deserialized data that is received.

Control: Sets the transmit and receive bit field size, bit sequence, clock polarity for SPI, and clock edge for SPI.

Status: Provides indication of transmit complete, receive complete, and any errors detected.

Errors include:

- Baud rate error (peripheral mode only): If the incoming clock exceeds 100%, the baud rate error becomes set.
- Rx data violation: If the previous receiving value is overwritten without being read by a bus controller, the Rx data violation becomes set.
- Frame size error (SPI mode only): If the received data size differs from the programmed value, a frame error is indicated.

CURRENT ACQUISITION UNIT (CAU)

The current acquisition unit (CAU) is a three-channel analog-to-digital converter (ADC) unit with a single current-sense amplifier dedicated to phase-current sensing using a single low-side shunt resistor. The unit, shown in functional form in Figure 7, includes a programmable-gain current-sense amplifier with three independent conversion channels, one for each phase. Each conversion channel has a fast sample-and-hold (S/H) circuit and a 12-bit, 1 μ s ADC. The fast capture-and-convert time allows the maximum time for a motor-control algorithm to calculate the field-oriented control (FOC) algorithm before the next PWM duty cycle.

An additional programmable-gain amplifier provides a single ground referenced voltage for additional external processing, if required.

Each channel in the CAU can be independently triggered by one of four sources: PWM generator, timers 1 through 4, GPIO, or software. The trigger edge is selectable as the rising edge, the falling edge, or both. Use of the PWM generator as the trigger source permits synchronization of the capture-and-conversion feature with the PWM switching edges under full control of the PWM generator block. The timing of the ADC trigger relative to the PWM edge is configured in and controlled by the PWM generator unit. The three-channel capture system triggered by the PWM generator permits three-phase current measurement using a single current-shunt resistor.

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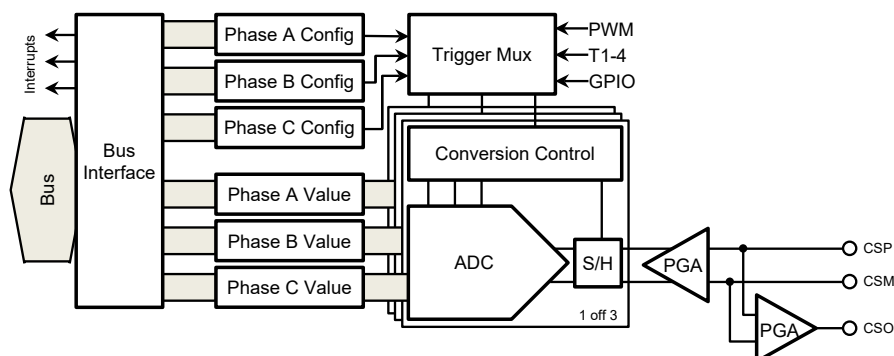


Figure 7: CAU Block Diagram

Conversion Sequence

When a trigger is received, the ADC conversion control first generates a signal for the sample-and-hold to capture the signal from the current-sense amplifier, then sequences the ADC through the conversion. The sample time and the conversion time are individually programmable in 4 steps from 1 μ s to 2.5 μ s. Upon completion of the conversion, the data is transferred to the output buffer, a status bit is set, and (if enabled) an interrupt request signal is generated. The conversion sequence is shown in Figure 8.

The conversion result can be handled in one of two ways.

- If a result-overwrite is permitted, the previous result is overwritten upon completion of a new conversion, and a CAU error is indicated.
- If a result-overwrite is not permitted, a conversion request is ignored until the previous result is read. If a conversion is requested before the data is read, a CAU error is indicated.

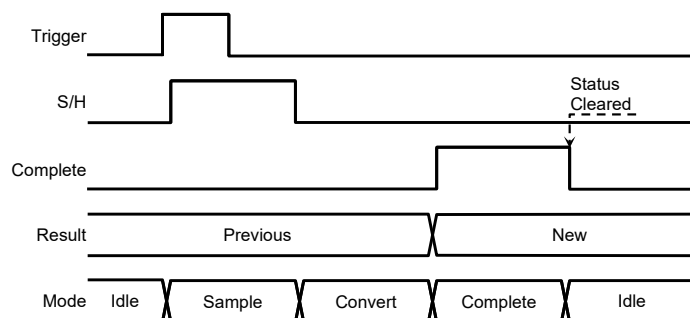


Figure 8: CAU Sequence

DIFFERENTIAL CURRENT-SENSE AMPLIFIER

The programmable-gain differential sense amplifier allows the use of a low-value current-shunt sense resistor as a current-sensing element in the low-side connection between the common-source node of all phases and the power ground return. The input common-mode range of the CSP and CSM inputs allows below-ground current sensing, which is typically required for low-side current sense in PWM control of motors, or other inductive loads, during switching transients. The gain can be set to optimize shunt power dissipation, peak current measurement capability, and current accuracy for the selected application. The connection from the sense amplifier to the ADC is maintained as a differential signal to allow maximum dynamic range and minimum noise in the analog signal path.

The available gain settings (DAG) and resulting maximum differential input voltage, V_{IDMAX} are listed in Table 2.

CSO CURRENT SENSE AMPLIFIER

The additional programmable -gain amplifier provides a single ground-referenced voltage output at the CSO terminal with programmable pedestal voltage and independent programmable gain. The output of this amplifier is supplied by the VLR regulator. This limits the maximum output to not greater than V_{LR} . The offset of this amplifier can be programmed to one of eight levels as listed in Table 2.

The gain of the CSO amplifier can be programmed independently from the differential amplifier; however, for operations with V_{LR} at 5 V, settings are provided to match the differential amplifier maximum differential input voltage. Gain settings and resulting maximum differential input voltage at listed in Table 2.

Equations describing the relationship between sense-amplifier gain and pedestal voltage are shown in Figure 9. The pedestal voltage value can be selected from 0 to 2.4 V.

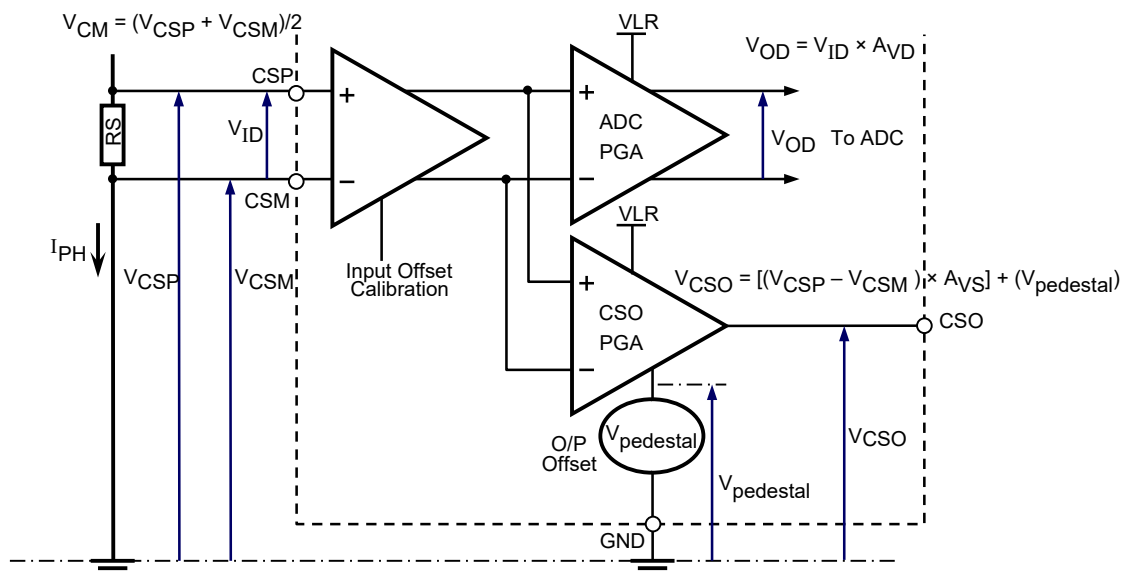
Table 2: Amplifier Gain and Offset Settings

DAG [1]	SAG [2]	V _{IDMAX} [3] (mV)
5	10	±240
8	16	±150
10	20	±120
13	26	±92
15	30	±80
18	36	±67
20	40	±60
25	50	±48

[1] DAG: Differential amplifier gain

[2] SAG: CSO amplifier gain

[3] V_{IDMAX}: Resulting maximum differential input voltage



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DIFFERENTIAL INPUT AMPLIFIER

The CSO amplifier includes a software-controlled input offset calibration feature which will ensure an input offset voltage, VIOS, of better than $\pm 1\text{mV}$. Unless calibrated, input offset voltage may exceed the limits detailed in the Electrical Characteristics table. Calibration is not required to achieve the specified Input Offset Voltage Drift, ΔVIOS , and leaves this parameter unaltered. Before initiating a calibration, the positive and negative amplifier input terminals (CSP and CSM) must be held at the same potential by ensuring that no current is flowing in the associated sense resistor and this condition must be maintained until the calibration operation is complete.

DATA ACQUISITION UNIT (DAU)

The data acquisition unit (DAU) has a 16-channel analog multiplexer, fast sample-and-hold (S/H), and a 12-bit analog-to-digital converter (ADC) with a programmable conversion time from $1\text{ }\mu\text{s}$ to $2.5\text{ }\mu\text{s}$. A block diagram of the DAU is shown in Figure 10.

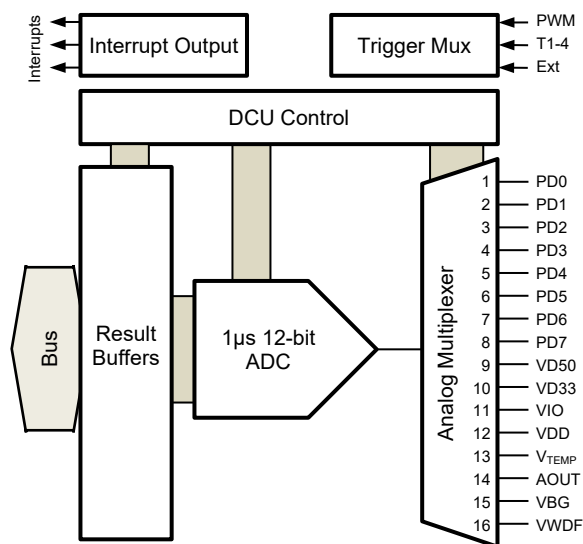


Figure 10: DAU Block Diagram

Input Sources

The DAU accepts 16 analog inputs, three of which are reserved for system diagnostic use. The inputs include the eight GPIO terminals plus three core voltages to monitor the supply, reference, and processor temperature. The AOUT input is a special analog input from the power management and gate drive section of the A89201: It allows selection of system and output voltage measurements as shown in Table 3.

Table 3: AOUT Signal Selection

Selection	Signal
0	VBB
1	VBRG
2	SA
3	SB
4	SC
5	VREG
6	VCP
7	VLR
8	VTJ
9	IG
10	OOS
11	BIASRB
12 – 15	GND

Conversion Source Selection

The DAU can be programmed to operate in three modes: single source, triggered sequence, and sequential trigger. In all modes, the sampling time for each channel to be converted can be independently programmed from $1\text{ }\mu\text{s}$ to $625\text{ }\mu\text{s}$.

In single-source mode, only one channel is selected for sampling and conversion. This selection can change at any time; however, if the selection is changed while a conversion is in progress, the new configuration only takes effect after the conversion is complete.

For the triggered-sequence and sequential-trigger modes, a sequence of up to 16 channels can be entered to be sampled and converted. Channels can be repeated in the sequence, and the sequence can be reduced in length by disabling specific samples. An example is shown in Table 4.

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Table 4: Channel Sequence Example

Sequence Register	Channel	Sample Time	Result Register
0	AOUT	1 μ s	0
1	VTEMP	5 μ s	1
2	PD4	1 μ s	2
3	VBG	1 μ s	3
4	PD0	1 μ s	4
5	Disabled	—	—
6	Disabled	—	—
7	GND	25 μ s	7
8	Disabled	—	—
9	Disabled	—	—
10	Disabled	—	—
11	Disabled	—	—
12	Disabled	—	—
13	VTEMP	5 μ s	13
14	Disabled	—	—
15	Disabled	—	—

In this example, only the signals at AOUT, VTEMP, PD4, VBG, PD0, and GND are sampled and converted. The sampling and conversion sequence starts with AOUT, followed by VTEMP, then PD4, VBG, and PD0; next, it jumps directly to GND, then to VTEMP again.

NOTE: The sequence is random, and signals can be selected more than once in the sequence.

In the triggered-sequence mode, the selected trigger starts a single sequential pass of all the selected channels. Each channel is sampled, then converted, in the order entered in the channel-sequence registers. The results of each conversion are then available in the result register associated with each channel sequence register.

In the sequential-trigger mode, the selected trigger starts a sample-and-conversion of the next sequential entry in the channel-

sequence register. The sequence continues through the channel-sequence register one channel at a time, upon each trigger input, rolling over from the last channel in the sequence to the first. For example, in the example in Table 4, following the second instance of the VTJ signal, the next signal to be sampled would be AOUT. The results of each conversion are then available in the result register associated with each channel-sequence register.

Trigger Source and Mode

The DAU can be triggered by one of four sources: PWM center, software, timer 1 through 4, or external (GPIO). For each, the trigger edge can be selected as rising, falling, or both, as shown in Figure 11.

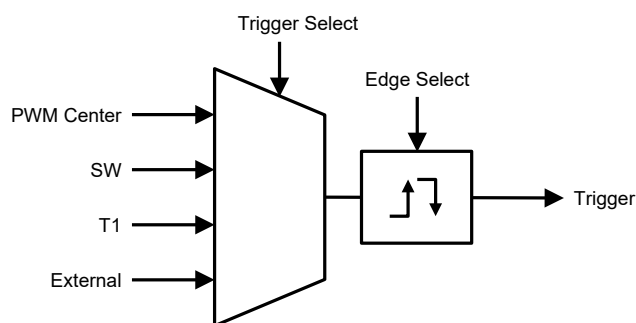


Figure 11: DAU Trigger-Select Options

Ignition Terminal (IG)

A89201 has an ignition terminal that is battery-voltage compliant. The terminal can be used to wake up the device or to call-interrupt on the CPU. The terminal has an internal passive pull-down resistor; therefore, an open circuit pulls the terminal to low.

When the device is in sleep mode, the device wakes up upon a low-to-high transition on the IG terminal.

When the device is active, a transition on the IG terminal can call the IRQ. For more details, refer to the [GDU user manual](#).^[1]

Sensor Bias Terminal (BIAS)

A89201 has a dedicated terminal for external sensor use. Two

^[1] This user manual is available for download from the Allegro Software Portal at <https://registration.allegromicro.com/admin/#/parts/A89201>. Registration is required.

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modes are available: current mode and voltage mode. A block diagram of current mode is shown in Figure 12. A block diagram of voltage mode is shown in Figure 13.

When current mode is selected, the BIAS terminal outputs a constant current, configurable from 25 μ A to 2.5 mA. The generated voltage on the terminal can be measured by DAU. For more detail, refer to the [GDU and DAU user manuals](#) available for download from the Allegro Software Portal. [1]

When voltage mode is selected, the BIAS terminal supplies a 5 V constant voltage. When this mode is active, external 2.2 μ F is required.

In the voltage mode, undervoltage detection can be active. If undervoltage is detected, the fault flag becomes active in SBOU in the GDU status register. When the MCU is configured, the GDU fault interrupt call is active, and automatic polling is active, the CPU automatically executes the GDU IRQ handler. In the

voltage mode, DAU access is available. The output voltage can be confirmed with the DAU feature.

For more detail, refer to the [GDU user manual](#) available for download from the Allegro Software Portal. [1]

If the BIAS terminal is not in use, it can be disabled. When it is disabled, the BIAS terminal becomes a high-impedance node. The BIAS function remains in sleep mode, which reduces internal power consumption.

The BIAS terminal is battery compliant. In the event of a short to the supply, the device does not become damaged.

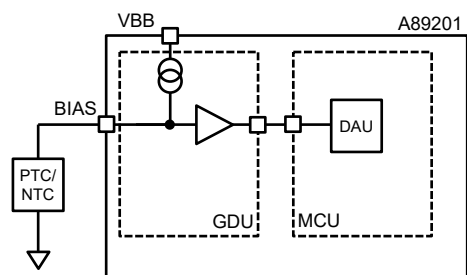


Figure 12: BIAS Terminal Block Diagram in Current Mode

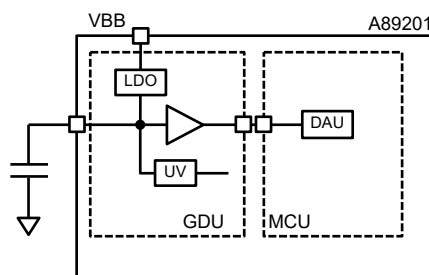


Figure 13: BIAS Terminal Block Diagram in Voltage Mode

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BEMF and Phase Comparator

BEMF COMPARATOR

A89201 has an integrated BEMF comparator that can be used to detect prerotation of the motor in a PMSM application. The A89201 can detect the frequency, rotation angle, and direction of the motor. This allows synchronization between the FOC and the prerotating speed, without causing any disturbance or noise. A detailed block diagram is shown in Figure 14. The AMCT block takes the BEMF signal and processes it accordingly. To connect the right signal to the AMCT, the GDU multiplexer needs to be configured. For more detail, refer to the [GDU user manual](#) available for download from the Allegro Software Portal. [1]

When the three-phase bridge is turned off, if the motor is rotating, the permanent magnet generates BEMF. The internal BEMF comparator creates a pulse synchronized with the rotation frequency.

PHASE COMPARATORS

A89201 has an internal fast phase comparator. This comparator is designed to compare the phase voltage to the half voltage VBRG. The block diagram is shown in Figure 15.

The phase comparator can be used to detect the phase-voltage phase for three-phase bridge verification. Also, this fast comparator can be used to detect current polarity while the bridge is in the dead-time period.

LIN Interface Module

The LIN Interface module provides the basic time-critical transmit and receive functions required to implement a LIN peripheral node. The module includes a LIN physical interface, asynchronous receiver, transmit serializer, LIN framing, synchronization and message identifier (PID) extraction. Message PID recognition and response generation are managed by the CPU.

The module automatically synchronizes to any incoming frames and detects any errors in transmission or invalid data in any frame. It then extracts the necessary data for the CPU to process. Data provided by the CPU for transmission is sent in the correct frame format, and a valid checksum for the data sent is appended.

Features

- ISO17987/J2602 compliant
- LIN 1.3 and 2.2x compatible
- LIN protocol implementation at data-link layer; processing complete LIN message frames
- Transmission rates from 1 kHz to 20 kHz
- Full LIN message buffering of identifier and 9 data bytes
- Automatic bit-rate detection and frame synchronization
- Wake on LIN

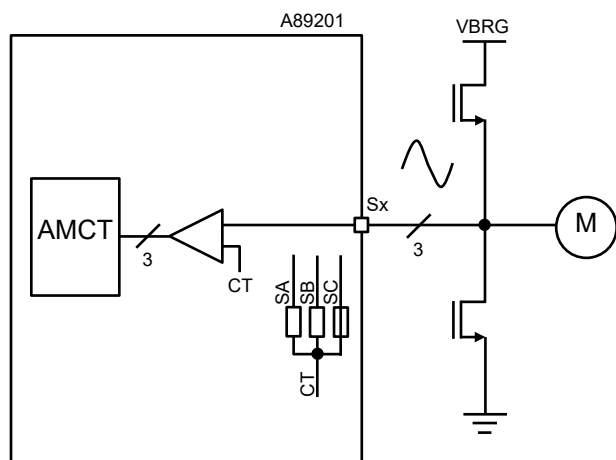


Figure 14: BEMF Comparator Block Diagram

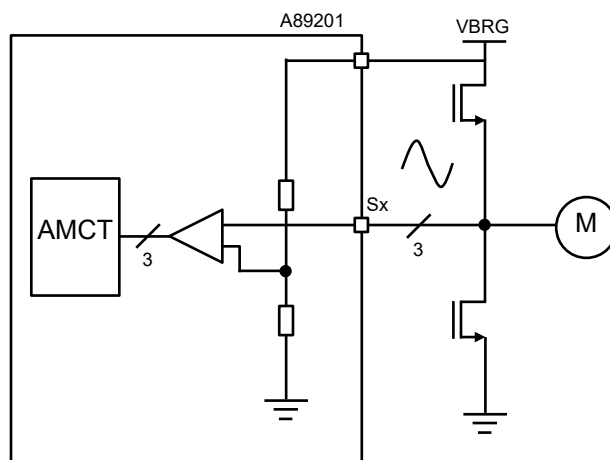


Figure 15: BEMF Comparator Block Diagram

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- Supports peripheral wake request
- Automatic bus idle timeout detection
- Comprehensive error detection
- Provides indicators for:
 - Frame header received
 - Transmit/Receive data complete
 - Wakeup detect
- Support for classic (LIN 1.3) checksum and enhanced (LIN 2.0) checksum
- Independent receive and transmit data paths
- Allows a clock deviation of up to 14% between the peripheral and controller oscillators

The functional diagram of the LIN interface module is shown in Figure 16. The asynchronous receiver (AR) and the frame processor monitor the LIN bus for any incoming sync break/sync field, which identifies the transmission of a new frame. The frame processor also detects the baud rate of the incoming frame and automatically synchronizes the receiver to the LIN baud rate.

After synchronization and error checking, the frame processor extracts the PID and passes any received data to the receive data buffer. Multiple registers are provided for received bytes and count, transmit bytes and number, message PID, communication status, and module configuration. The frame processor also monitors and handles errors in the received data stream. If an error is detected, the current receive operation terminates, and the frame processor waits for a new frame header.

When data is written to the transmit data buffer and a byte count is written to the configuration register, the data bytes are assembled into the correct frame format by the framer and are sent to the LIN bus through the serializer.

This module monitors the transmitted data by using the receive path to ensure that the transmit and receive data match. If they do not match, for example due to another peripheral or controller transmitting at the same time, an error is indicated and the transmission is terminated. The module also detects any timeout errors that occur on the LIN bus and indicates if a frame response is not received or if the transmit path is locked in the dominant state.

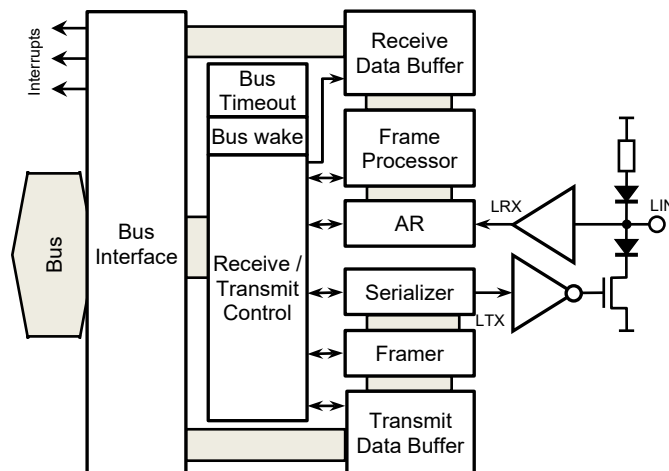


Figure 16: LIN Interface Module Block Diagram

LIN PHYSICAL INTERFACE

The LIN physical interface drives and monitors a single-wire LIN bus as a peripheral node that complies with the ISO17987 (LIN 2.x) and SAE J2602 standards. The LIN terminal can withstand voltages from -40 V to 50 V with respect to the ground pin without adversely affecting LIN bus communications between other devices.

The LIN terminal meets all the voltage, timing and slew limitation requirements of LIN 2.2 when actively transmitting and when receiving. A timer is included to ensure that the LIN output does not remain dominant if a fault occurs. If the LIN terminal is driven in the dominant state for longer than the transmit-dominant timeout period, t_{TXTO} , the output is disabled and allowed to return to the recessive state to avoid blocking the LIN bus for other messages.

Data to be transmitted is passed from the serializer (Figure 12) to the LIN interface through the LTX signal and converted to LIN bus signals. A logic high on the LTX signal produces a recessive bus (high) state; a logic low produces a dominant bus (low) state.

The logic state of the LIN bus is determined by the receiver and is passed to the asynchronous receiver (AR; see Figure 12) through the LRX signal. When the LIN bus is in the dominant

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(low) state, LRX is low; when the LIN bus is in the recessive (high) state, LRX is high. In the sleep state, LRX is not active and is low.

The LIN terminal can also be used as a control PWM input by selecting the LRX signal as an input to one of the timers. This allows the duty cycle of the controlling PWM signal to be interpreted by the CPU and the appropriate action to be taken.

LIN Wake

When the A89201 is in the minimum-power sleep state, the LIN terminal changes to a passive input and the resistance of the pull-up resistor on the LIN terminal increases to approximately 2 MΩ. This ensures that the LIN terminal is unable to affect the LIN bus signal. In the minimum-power sleep state, the LIN terminal continues to be monitored by a micropower circuit in order to detect a wake request.

Two wake sequence modes are possible, determined by the selected operating mode:

- In LIN mode, the LIN terminal complies with the requirements of the LIN standard regarding dominant timeout and wake on LIN.
- In PWM mode, the LIN dominant timeout is disabled and the wake mode is changed to wake on a simple rising or falling edge.

LIN Mode

In LIN mode the A89201 wakes up according to the LIN proto-

col. In this mode, the wake request is valid on the first low-to-high transition on the LIN terminal after the LIN terminal is in a dominant (low) state for longer than the wake time, t_{BUSWK} , as shown in Figure 17. If the LIN terminal changes to recessive (high) within t_{BUSWK} , the A89201 returns to the sleep state.

When a valid wake request is detected on the LIN terminal, or when the IG terminal transitions from low to high, the A89201 exits the typical sleep state, turns on all regulators and control circuits, and commences operation.

The LIN interface can be controlled by the CPU to remain in the standby state, where the LIN terminal is a passive input. The LRX output indicates the state of the LIN terminal; however, the signal on the LTX terminal is ignored.

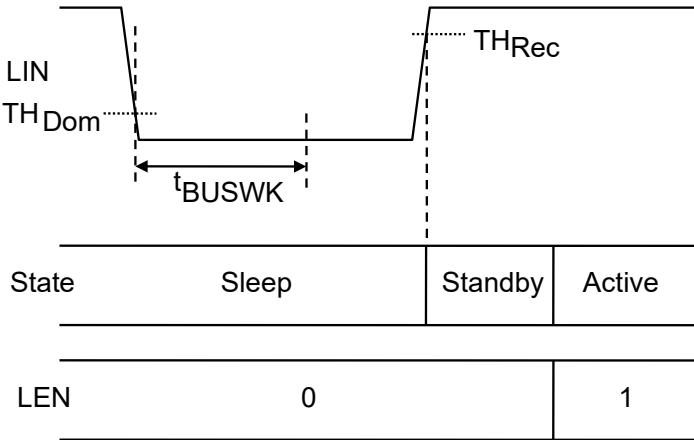


Figure 17: LIN Wake Sequence and Timing

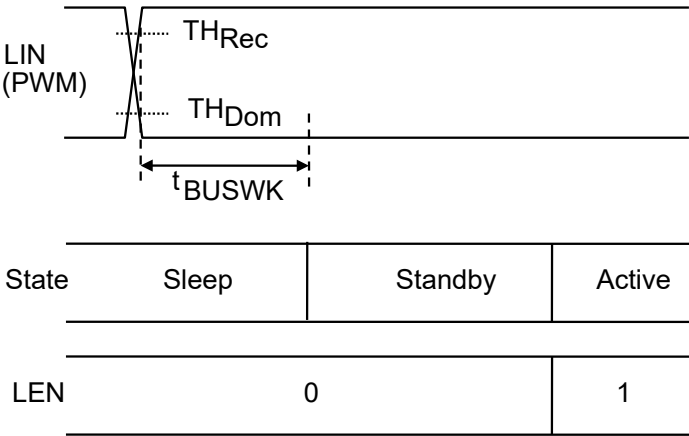


Figure 18: PWM Wake Sequence and Timing

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PWM GENERATOR (PGU)

The PWM generator unit contains three PWM generators, one for each phase in a four-phase system. Each generator provides three outputs: a high-side gate drive signal, GH; a low-side gate drive signal, GL; and two current-sample triggers to the CAU to sample the current-sense signal at a specific time during the PWM period for each phase. The PWM switching points for each phase are updated by the CPU upon each PWM period or upon a multiple of the PWM period.

The controller clock, f_{PWM} , from the CCU, drives a timer that synchronizes the signals across all three generators. It also drives a PWM center identifier circuit that generates a 50% duty cycle signal with a rising edge at the center of the PWM period and a falling edge at the start of the PWM period.

The PGU has the following features:

- Three PWM generators
- Two ADC trigger signals from each generator
- Two PWM outputs from each generator
- Independent control of PWM outputs
- Center or edge-aligned modulation
- Programmable dead time
- 12.5 ns PWM resolution
- PWM frequency of 4.9 kHz up to 5 MHz
- Dead-time generation of up to 12.7875 μ s
- Phase shift in center-aligned and edge-aligned modes

All timing and frequencies assume 80 MHz f_{PWM} clock.

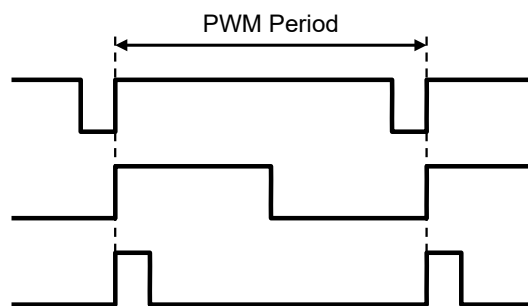


Figure 19: Edge-Aligned PWM

MODULATION

Two modulation modes are supported by the PGU; edge-aligned and center-aligned. An example of edge-aligned is shown in Figure 19, and an example of center-aligned is shown in Figure 20. Center-aligned mode is needed to operate with a single-shunt FOC algorithm; however, the edge-aligned mode is provided as an option for block commutation (trapezoidal drive) control algorithms. Edge-aligned mode is also useful for initial-position detection algorithms. The selected modulation can be changed each PWM period.

CURRENT-SAMPLE TRIGGER POINT CONTROL

The PWM generator has dedicated programmable outputs to control the current measurement points. This feature allows performance of three-phase current measurement using a single-shunt resistor. A fine delay control depending on the bridge design is also possible.

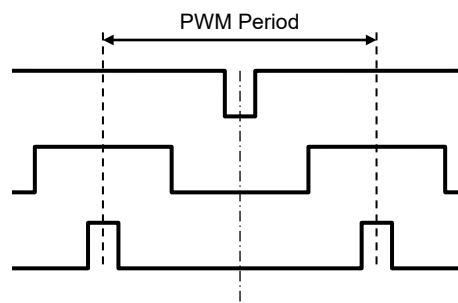


Figure 20: Center-Aligned PWM

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Advanced Motor Control Timer (AMCT)

The AMCT provides the necessary timers and functions to offload time-critical time measurements from the CPU. The unit provides four basic subsystems: one for Hall sensor-based applications, one for sensorless BEMF-based applications, one with features to assist with sensorless initial-position detection, and one for current polarity detection.

In summary, the features provided are:

- Angle error report with respect to the BEMF zero-crossing point
- Electrical motor speed in a Hall-effect-based system
- Phase advance and delay application in both BEMF and Hall-effect-based control
- Multistage filters for BEMF comparators to remove PWM switching noise and coil degaussing effect
- Filters for Hall sensors to avoid the switching noise
- Calculation of commutation state for trapezoidal drive and estimated motor angle for sinusoidal drive
- Automatic disabling of the bridge phases depending on the commutation section for high-speed applications
- Initial-position detection of motor within $\pm 15^\circ$ elec.

HALL-BASED SUBSYSTEM

Three main tasks performed by this part of the AMCT are a Hall sensor debounce filter, a phase-shift generator to achieve phase advance and delay, and a Hall-state-change timer to permit motor-frequency calculation.

The debounce filter is used mainly to avoid the problems associated with PWM switching noise. If the Hall sensor outputs are noisy due to a PWM switching condition, this filter can be used to remove the noise.

- The phase-shift generator provides an electrical angle (time) shifted version of the Hall sensor inputs as commutation states: For phase advance, the CPU provides a negative value
- For phase delay, the CPU provides a positive value.

The Hall frequency calculator uses one timer to count the number of clock cycles elapsed between two different Hall state inputs. When the new Hall state appears, the counter value is latched and is available to the CPU to calculate the Hall commutation frequency.

SENSORLESS BEMF-BASED SUBSYSTEM

This part of the AMCT provides filtering stages for the BEMF comparators and a number of calculations relating to the BEMF motor angle estimation. These include trapezoidal commutation state, estimated motor electrical position, angle error with respect to BEMF zero-crossing point, and a filtered version of BEMF comparator outputs.

The filter section includes three programmable filters, each with a specific function. The BEMF comparator filter removes problems associated with switching noise to provide a clean comparison result. This is followed by a debounce filter that removes any effects caused by the noise associated with PWM switching. Finally, a degaussing filter can be enabled in order to remove the effects on the BEMF caused by the decaying current remaining in the inactive phase.

The BEMF comparator output is also used to generate a speed and direction signal that can be used to detect if a motor is rotating (windmilling) before attempting to start the motor. Depending on the motor speed and direction, this data can be used to determine if a synchronous start or a brake and restart should be used.

INITIAL POSITION DETECTION (IPD)

Several programmable features are included that enable the initial rotor position to be estimated to within $\pm 15^\circ$ elec. Short-duration low-level current pulses are used to measure the relative inductance in each phase. The relative inductance in each phase is used to determine the rotor position. The rotor position is used to align the phase drive with the motor and to start the motor without pre-alignment or reverse rotation.

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Gate-Drive Unit (GDU)

The gate-drive outputs of the A89201 are provided by the GDU and controlled directly by the PWM generator unit. These outputs are designed to drive external, low-on-resistance, power N-channel MOSFETs. Each drive supplies the large transient currents necessary to quickly charge and discharge the external MOSFET gate capacitance to reduce dissipation in the external MOSFET during switching. The charge current for the low-side drives and the recharge current for the bootstrap capacitors is provided by the capacitor on the VREG terminal. The charge current for the high-side drives is provided by the bootstrap capacitors connected between the Cx and Sx terminal, one for each phase. MOSFET gate charge and discharge rates may be controlled by setting a group of parameters in the drive control registers.

BOOTSTRAP SUPPLY

When the high-side drivers are active, the reference voltage for the gate-drive output rises to nearly that of the bridge supply voltage. To ensure that it remains active, the supply to the gate drive must then exceed the bridge supply voltage. This high-side supply is provided by bootstrap capacitors, one for each high-side driver. These three bootstrap capacitors, connected between the bootstrap supply terminals (CA, CB, and CC) and the corresponding high-side reference terminals (SA, SB, and SC), are necessary to provide the initial gate-drive current pulse necessary to allow fast MOSFET switching for the lowest switching losses in the MOSFETs.

When the associated reference Sx terminal is low, the bootstrap capacitors are independently charged to approximately V_{REG} . When the output swings high, the voltage on the bootstrap supply terminal increases with the output to provide the boosted gate voltage needed for the high-side N-channel power MOSFETs.

When the PWM duty cycle is high, the Sx terminal remains high for extended periods of time. The gate drive voltage is maintained by an additional charge pump that provides a regulated voltage to charge the bootstrap capacitor. This charge pump also provides the charging current when the output PWM duty cycle is too high to permit the bootstrap capacitor to be recharged between high-side turn-on events. This combination of bootstrap drive and charge pump provides the most efficient gate-drive system for a PWM system able to operate between 0 and 100% without restriction.

HIGH-SIDE GATE DRIVE

High-side, gate-drive outputs for external N-channel MOSFETs are provided on terminals GHA, GHB, and GHC:

- $GHx = 1$ (or “high”) means that the upper half of the driver is turned on and its drain sources current to the gate of the high-side MOSFET in the external motor-driving bridge, turning it on.
- $GHx = 0$ (or “low”) means that the lower half of the driver is turned on and its drain will sink current from the external MOSFET’s gate circuit to the respective Sx terminal, turning it off.

The Sx terminals are connected to the negative side of the bootstrap capacitors and are the negative supply reference connections for the floating high-side drivers. The discharge current from the high-side MOSFET gate capacitance flows through these connections, which should have low-impedance traces to the MOSFET bridge.

LOW-SIDE GATE DRIVE

The low-side, gate-drive outputs on GLA, GLB, and GLC are referenced to the LSS terminal. These outputs are designed to drive external N-channel power MOSFETs:

- $GLx = 1$ (or “high”) means that the upper half of the driver is turned on and its drain will source current to the gate of the low-side MOSFET in the external power bridge, turning it on.
- $GLx = 0$ (or “low”) means that the lower half of the driver is turned on and its drain sinks current from the external MOSFET gate circuit to the LSS terminal, turning it off.

The LSS terminal provides the return path for discharge of the capacitance on the low-side MOSFET gates. These terminals are connected independently to the common sources of the low-side external MOSFETs through a low-impedance track.

Both high-side and low-side MOSFET gate charge and discharge rates may be controlled as described next, via the registers, to provide slew-rate control or lowest-loss fast switching.

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BOOTSTRAP CAPACITOR SELECTION

The A89201 requires three bootstrap capacitors: CA, CB, and CC. To simplify the description of bootstrap capacitor selection criteria, generic naming is used here. For example, C_{BC} , Q_{BC} , and V_{BC} refer to any of the three capacitors above, and Q_{GATE} refers to any of the six associated MOSFETs.

To ensure proper operation of the device, C_{BC} must be selected correctly:

- A large value results in extra time required to charge the capacitor and limits the achievable peak duty cycle and frequency.
- A small value results in extra voltage drop when charge is transferred from C_{BC} to the MOSFET gate.

To minimize the voltage drop due to charging, the charge in the bootstrap capacitor, Q_{BC} , should be much larger than Q_{GATE} , which is the charge required by the gate:

A factor of 20 is a reasonable value, so:

$$Q_{BC} = C_{BC} \times V_{BC} = Q_{GATE} \times 20; \text{ or}$$

$$C_{BC} = Q_{GATE} \times 20 / V_{BC},$$

where V_{BC} is the voltage across the bootstrap capacitor.

The voltage drop, ΔV , across the bootstrap capacitor as the MOSFET is being turned on can be approximated by:

$$\Delta V = Q_{GATE} / C_{BC}$$

Therefore, for a factor of 20, ΔV is 5% of V_{BC} .

The maximum voltage across the bootstrap capacitor for typical operating conditions is $V_{REG(max)}$. However, in some circumstances, the voltage may reach a maximum of 18 V due to transient conditions of the clamp voltage of the Zener diode between the Cx terminal and the Sx terminal. In most applications with a good ceramic capacitor, the working voltage can be limited to 16 V.

VREG CAPACITOR SELECTION

The internal reference, V_{REG} , supplies current for the low-side gate-drive circuits and the charging current for the bootstrap capacitors. When a low-side MOSFET is turned on, the gate-drive circuit provides high-transient current to the gate. This high-transient current is necessary for quick turn-on of the MOSFET. This current, which can be several hundreds of milliamperes, cannot be provided directly by the limited output of the V_{REG} regulator, but must be supplied by an external capacitor, CVREG, connected between the VREG terminal and GND.

The turn-on current for the high-side MOSFET is similar in value but is mainly supplied by the bootstrap capacitor. However, the bootstrap capacitor must then be recharged from C_{VREG} through the VREG terminal. Unfortunately, the bootstrap recharge can occur a very short time after the low-side turn-on occurs. This means that the value of C_{VREG} between V_{REG} and GND should be high enough to minimize the transient voltage drop on V_{REG} for the combination of a low-side MOSFET turn-on and a bootstrap capacitor recharge.

For block commutation control (trapezoidal drive), where only one high side and one low side switch during each PWM period, a minimum value of $20 \times C_{BC}$ is reasonable. For sinusoidal control schemes, a minimum value of $40 \times C_{BC}$ is recommended. Because the maximum working voltage of C_{VREG} never exceeds V_{REG} , the capacitor voltage rating can be as low as 15 V. However, to reduce any impact operating voltage may have on the capacitance value, a capacitor rated to at least twice the maximum working voltage is recommended. For the best performance, CVREG should be ceramic rather than electrolytic. CVREG should be mounted as close to the VREG terminal as possible.

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Gate Drive, Drive-Strength Control

MOSFET gate drives are controlled according to the values set in the slew-fall and slew-rise registers.

High-side off-to-on transitions are controlled as detailed in Figure 21. When a gate drive is commanded to turn on, a current, I_1 , is sourced on the respective GHx terminal for a duration, t_1 . Because drain-source voltage does not change during this period, these parameters should typically be set to quickly charge the MOSFET gate-source capacitance to the start of the Miller region. At the end of t_1 the current sourced on GHx is set to a value of I_2 and remains at this value while the MOSFET transitions through the Miller region and reaches the fully-on state, defined as:

$$V_{DS} = (V_{BRG} - V_{SX}) < V_{DSTH}$$

I_2 should be set to achieve the required gate-drain capacitance discharge time. Once in the fully-on state, the GHx output switches from current drive to voltage drive to hold the MOSFET in the on state.

If the slew currents are set to 0, GHx produces the maximum drive to turn on the MOSFET as quickly as possible without attempting to control the MOSFET capacitance charge time (Figure 22). The switching-time setting does not affect switching speed.

Low-side off-to-on transitions are controlled in a similar manner by setting the slew currents and timing to control GLx, with fully-off defined as:

$$V_{DS} = (V_{SX} - V_{LSSX}) < V_{DSTL}$$

For the low-side, I_2 should be set to achieve the required falling slew time on the motor phase connection.

High-side on-to-off transitions are controlled as detailed in Figure 23. When a gate drive is commanded to turn off, a current, I_1 , is sunk by the relevant GHx terminal for a duration, t_1 . Because drain-source voltage does not change during this period, these parameters should typically be set to quickly discharge the MOSFET gate-source capacitance to the start of the Miller region. At the end of t_1 , the current sunk by GHx is set to a value of I_2 and remains at this value while the MOSFET transitions through the Miller region and reaches the fully-off state, defined as:

$$V_{DS} = (V_{SX} - V_{LSSX}) < V_{DSTH}$$

I_2 should be set to achieve the required gate-drain capacitance charge time. Once in the fully-off condition, the GHx output switches from current drive to voltage drive to hold the MOSFET in the off state.

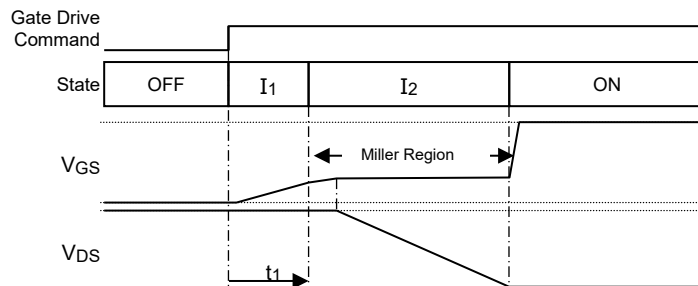


Figure 21: Off-to-on Transition (Gate Drive Control)

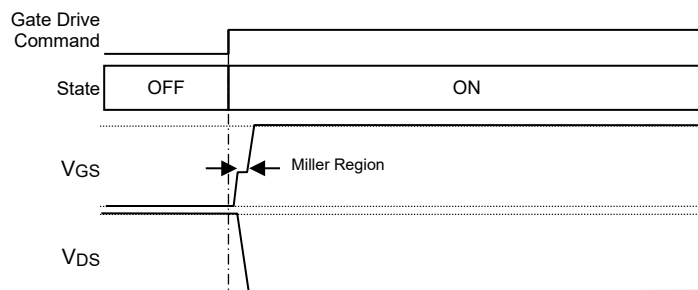


Figure 22: Off-to-on Transition (Switched)

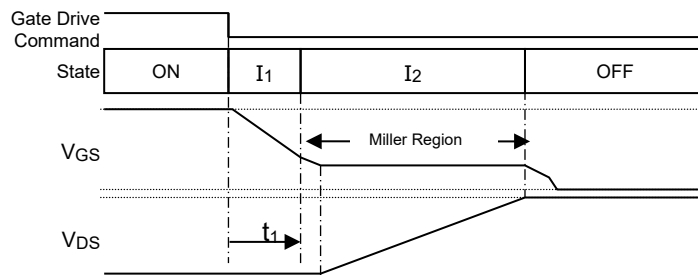


Figure 23: On-to-off Transition (Gate Drive Control)

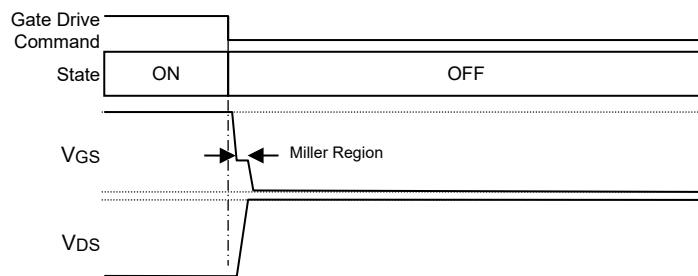


Figure 24: On-to-off Transition (Switched)

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If the slew currents are set to 0, GLx produces maximum drive to turn off the MOSFET as quickly as possible without attempting to control the input capacitance discharge time (see Figure 24). The switching-time setting does not affect switching speed.

Low-side on-to-off transitions are controlled in a similar manner by setting the slew currents and timing to control GLx and the fully-off state, defined as:

$$V_{DS} = (V_{BRG} - V_{SX}) < V_{DSTH}$$

For the low side, I_2 should be set to achieve the required rising slew time on the motor phase connection.

DEAD-TIME SETTING

When using the controlled drive facility, the dead time programmed into the PWM generator must be long enough to ensure any MOSFET turning off fully transitions to the off state before the complementary MOSFET in the same phase is allowed to start turning on.

GDU DIAGNOSTICS

Diagnostic Monitors

Multiple diagnostic features provide three levels of fault monitoring. These include critical protection for the gate drives, monitors for operational voltages and states, and detection of power bridge and load fault conditions. All diagnostics, except for POR and EEPROM error, can be masked by setting the appropriate bit in the mask register. The fault status is available from the diagnostic and status registers.

Diagnostic Registers

Detailed diagnostic data can be read from the diagnostic and status registers at any time. The driver status register provides a summary of all faults in a single read transaction. The MSB (bit [15]) of the status register contains a common fault flag, FF. If a fault bit in the status register is set, FF is high. This allows the fault condition to be detected by simply polling the status register and checking the MSB. In all cases, the fault bits in the diagnostic and status registers are latched and are only cleared after a register reset.

Table 5: Diagnostic Functions

Name	Diagnostic	Level
POR	Internal logic supply undervoltage causing power-on reset	Chip
EE	EEPROM error	Chip
OT	Chip junction overtemperature	Chip
TW	High chip junction temperature warning	Monitor
ETO	Enable watchdog timeout	Monitor
VLU	VLR output undervoltage	Monitor
VRU	VREG output undervoltage	Monitor
VPU	VCP output undervoltage	Monitor
VA	Bootstrap undervoltage phase A	Bridge
VB	Bootstrap undervoltage phase B	Bridge
VC	Bootstrap undervoltage phase C	Bridge
AHO	Phase A high-side VDS overvoltage	Bridge
ALO	Phase A low-side VDS overvoltage	Bridge
BHO	Phase B high-side VDS overvoltage	Bridge
BLO	Phase B low-side VDS overvoltage	Bridge
CHO	Phase C high-side VDS overvoltage	Bridge
CLO	Phase C low-side VDS overvoltage	Bridge

Chip-Level Protection

Chip-wide parameters critical to the correct operation of the drive unit are monitored. These include maximum chip temperature, minimum internal logic supply voltage, and serial interface transmission. These three monitors are necessary to ensure that the GDU is able to respond as specified.

Chip Fault State: Internal Logic Undervoltage

The GDU has an independent integrated logic regulator to supply the internal logic. This is to ensure that external events, other than loss of supply, do not prevent the gate drives from operating safely. The internal logic supply regulator continues to operate with a low supply voltage, for example, if the main supply voltage drops to a very low value during transient load event. In extreme low-supply-voltage circumstances, or during power-up or power-down, an undervoltage detector ensures that the GDU

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operates correctly. The logic supply undervoltage lockout cannot be masked because it is essential to guarantee correct operation over the full supply range.

When power is first applied to the A89201, the internal logic in the GDU is prevented from operating, and all gate-drive outputs are held in the off state until the internal regulator voltage, V_{D50} , exceeds the logic supply undervoltage lockout rising (turn-on) threshold. This threshold is derived from the sum of the VBB POR threshold, V_{BBR} , and the VBB POR hysteresis, V_{BBRHys} . At this point, all serial control registers reset to their power-on state and all fault states reset. The FF bit and the POR bit in the status register set to one to indicate that a power-on-reset has occurred. The GDU then enters its fully operational state and begins to operate as specified.

Once the GDU is operational, the internal logic supply continues to be monitored. If, during the operational state, V_{D50} reduces to less than the logic supply undervoltage lockout falling (turn-off) threshold, derived from V_{BBR} , the logical function of the GDU cannot be guaranteed and the outputs are immediately disabled. The GDU enters a power-down state and all internal activity, other than the logic regulator voltage monitor, is suspended. If the logic supply undervoltage is a transient event, the GDU follows the previously described power-up sequence as the voltage increases.

Chip Fault State: Overtemperature

If the chip temperature increases to greater than the overtemperature threshold, T_{JF} , the overtemperature bit, OT, becomes set in the status register. If FOT = 1 when an over temperature is detected, all gate-drive outputs are disabled automatically. If FOT = 0, circuitry is not disabled and action must be taken by the user to limit the power dissipation in some way so as to prevent overtemperature damage to the chip and unpredictable device operation. When the temperature reduces to less than T_{JF} by more than the hysteresis value, T_{JFHys} , the outputs reactivate, but the overtemperature bit remains in the status register.

Chip Fault State: EEPROM

Configuration and calibration data is stored within internal EEPROM and is loaded into working registers to configure the device at power-up. As part of this process, a data integrity check occurs. If the check returns a single-bit error, automatic error correction is applied and the part starts up. If the check returns a multiple-bit error, all gate drives are disabled and the EEPROM error bit, EE, is set in the status register. EEPROM faults can only be cleared by a power-on reset (POR).

OPERATIONAL MONITORS

Parameters related to the safe operation of the GDU are monitored. These include parameters associated with external active and passive components and the power supply regulators.

Voltages relating to driving the external power MOSFETs are monitored, specifically: V_{REG} , each bootstrap capacitor voltage, and the V_{GS} of each gate-drive output. The main bridge voltage, V_{BRG} , is monitored for overvoltage and undervoltage events.

The battery-compliant logic inputs and outputs are capable of being shorted to the main supply voltage without damage; however, any high voltage on these terminals is detected.

Monitor: VREG Undervoltage

The internal charge-pump regulator supplies the low-side gate driver and the bootstrap charge current. It is critical to ensure that the regulated voltage, V_{REG} , at the VREG terminal is sufficiently high before enabling any of the outputs.

If V_{REG} reduces to less than the VREG undervoltage threshold, V_{ROFF} , the VREG undervoltage bit, VRU, is set in the status register.

If a VREG undervoltage state is present and FVRU = 1, all gate-drive outputs go low. When VREG increases to greater than the rising threshold, V_{RON} , the fault is cleared and, if FVRU = 1, the gate-drive outputs are re-enabled. The fault bit remains in the status register until cleared. If FVRU = 0, fault reporting is the same, but the gate-drive outputs are not disabled and appropriate action to avoid potential misoperation or damage to the GDU and/or bridge MOSFETs must be taken by the MCU.

The VREG undervoltage monitor circuit is active during power up. If FVRU = 1, all gate drives remain low until V_{REG} is greater than V_{RON} . If FVRU = 0, the gate-drive outputs are active as soon as there is sufficient voltage on VREG to activate the gate-drive outputs.

The VREG undervoltage monitor can be disabled by setting the VRU bit in the mask register. Although not recommended, setting VRU to 1 or setting FVRU to 0 can allow the GDU to operate at less than the minimum specified supply voltage level with a severely impaired gate drive. In this condition, the specified electrical parameters are not valid.

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Monitor: Temperature Warning

If the chip temperature increases to greater than the temperature warning threshold, T_{JW} , the hot warning bit, TW, becomes set in the status register; and, if FTW = 1, all gate drives are low. If FTW = 0, gate drives remain active. When the temperature reduces to less than T_{JW} by more than the hysteresis value, T_{JWHys} , the fault is reset; and, if FTW = 1, the outputs are re-enabled. The TW bit remains in the status register until reset.

Monitor: ENABLE Watchdog Timeout

The ENABLE input provides a direct connection to all gate-drive outputs and can be used as a safety override to immediately disable the outputs. The ENABLE input is programmed to operate as a direct logic control by default; however, it can be monitored by a watchdog timer by setting the ETO bit to 1. In the direct mode, the input is monitored; this is described in the Logic Control Inputs section of the [GDU user manual](#) available for download from the Allegro Software Portal. [1] In watchdog mode, the first change of state on the ENABLE input activates the gate-drive outputs under command from the corresponding phase-control signals, and a watchdog timer starts. The ENABLE input must then change state before the end of the ENABLE timeout period, t_{ETO} . If the ENABLE input does not change before the end of the timeout period, the ETO bit becomes set in the status register and, if FEN = 1, all gate-drive outputs are driven low (disabled). Any following change of state to the ENABLE input reactivates the gate-drive outputs. If FEN = 0, the faults are indicated and stored, but the gate-drive outputs are not affected. The ETO bit remains set in the status register until reset.

POWER BRIDGE AND LOAD FAULTS

Bridge: Bootstrap Capacitor Undervoltage Fault

The GDU attempts to ensure that the bootstrap capacitor remains charged at all times when it is in the operating state. The bootstrap capacitor voltage is monitored by an undervoltage comparator, which is active when the corresponding high-side gate drive is active (on). If the bootstrap voltage is less than the undervoltage threshold, V_{BCUV} , when the high-side is commanded on, or at any time while the high-side is on, a bootstrap undervoltage fault is detected.

When a valid bootstrap undervoltage fault is detected, the action taken and the fault-reset conditions depend on the state of the FVBU bit.

If FVBU = 0, the fault state is latched, the associated bootstrap undervoltage fault bit (VA, VB, VC) is set in the diagnostic 0 register, and the associated gate-drive output is inactive (low). The fault state is reset the next time the MOSFET is commanded to switch, but the bootstrap undervoltage fault bit remains. If the MOSFET is driven with a PWM signal, this typically means that the MOSFET turns on again each PWM cycle. If this is the case and the fault condition remains, a valid fault is again detected after the timeout period and the sequence repeats.

If FVBU = 1, the fault is latched, the associated bootstrap undervoltage fault bit (VA, VB, VC) is set in the diagnostic 0 register, and all gate-drive outputs are inactive (low). The gate-drive outputs remain inactive (low) until the fault state is reset.

The bootstrap undervoltage monitor can be disabled for all three phases by setting the BSU bit in the mask register. Although not recommended, this can allow the GDU to operate at less than its minimum specified supply voltage level with a severely impaired gate drive. The specified electrical parameters may not be valid in this condition.

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Bridge MOSFET VDS Overvoltage Fault

Faults on any external MOSFETs are determined by monitoring the drain-source voltage of the MOSFET and comparing it to a drain-source overvoltage threshold, V_{DST} . V_{DST} is generated by an internal DAC and is defined by the value in the VT [5:0] variables respectively. These variables provide the input to two 6-bit DACs with a typical least significant bit value of 50 mV. The output of the DAC produces the threshold voltage, approximately defined as:

$$V_{DST} = n \times 50 \text{ mV}$$

where n is a positive integer defined by VT [5:0]

The low-side drain-source voltage for any MOSFET is measured between the adjacent Sx terminal and the adjacent LSSx terminal. Using the LSS terminal rather than the ground connection avoids adding any low-side current sense voltage to the real low-side drain-source voltage and avoids false VDS fault detection.

The high-side drain-source voltage for any MOSFET is measured between the adjacent Sx terminal and the VBRG terminal. Using the VBRG terminal rather than VBB avoids adding any reverse-diode voltage or high-side current-sense voltage to the real high-side drain-source voltage and avoids false VDS fault detection.

The VBRG terminal is an independent low-current sense input to the top of the MOSFET bridge. It should be connected independently and directly to the common connection point for the drains of the power-bridge MOSFETs at the positive supply connection point in the bridge.

NOTE: The VBRG terminal can withstand negative voltage to -5 V. This allows the terminal to remain connected directly to the top of the power bridge during negative transients where the body diodes of the power MOSFETs are used to clamp the negative transient. The same applies to the more extreme case where the MOSFET body diodes are used to clamp a reverse-battery connection.

The output from each VDS overvoltage comparator is filtered by a VDS fault qualifier circuit. This circuit uses a timer to verify that the output from the comparator is indicating a valid VDS fault. The duration of the VDS fault qualifying timer, t_{VDQ} , is determined by the contents of the VDQ [5:0] VD33iable. t_{VDQ} is approximately defined as:

$$t_{VDQ} = n \times 100 \text{ ns}$$

where n is a positive integer defined by VDQ [5:0].

The qualifier can operate in debounce mode or blanking mode, as selected by the VDQ bit.

In debounce mode (the default setting) a timer is started each time the comparator output indicates a VDS fault detection when the corresponding MOSFET is active. This timer is reset when the comparator changes back to indicate typical operation. If the debounce timer reaches the end of the timeout period, set by t_{VDQ} , the VDS fault is considered valid and the corresponding VDS fault bit (ALO, AHO, BLO, BHO, CLO, or CHO) is set in the diagnostic 0 register.

In blanking mode (optional) a timer is started when any gate drive is turned on or turned off. The outputs from the VDS overvoltage comparators for all MOSFETs are ignored (blanked) for the duration of the timer active period, set by t_{VDQ} . If any gate drive changes state while a blanking period is in progress, the timer is re-triggered, resulting in an extended overall blanking time. If any comparator output indicates a VDS fault and the blanking timer is not active, the VDS fault is considered valid and the corresponding VDS fault bit (ALO, AHO, BLO, BHO, CLO, or CHO) is set in the diagnostic 0 register.

The action taken when a valid VDS fault is detected and the action then required to clear the fault state depend on the FDSO bit value.

If FDSO = 0, the fault state latches, the associated VDS fault bit sets, and the gate drive for the associated MOSFET becomes inactive (low). The gate-drive output remains inactive (low) until the fault state is reset the next time the MOSFET that detected the fault is commanded to switch to the on state. The associated VDS fault bit remains set in the diagnostic 0 register until reset. If the MOSFET is driven with a PWM signal, this typically means that the MOSFET turns on again each PWM cycle. If this is the case and the fault condition remains, a valid fault is again detected after the timeout period and the sequence repeats. The VDS fault bits remain until reset.

If FDSO = 1, the fault latches, the associated VDS fault bit sets, and all gate-drive outputs become inactive (low). The gate-drive outputs remain inactive (low) until the fault state is reset.

If FDSO = 0, care must be taken to avoid damage to the MOSFET where the VDS fault is detected. Although the MOSFET switches off as soon as the fault is detected at the end of the fault validation timeout, it is possible that the MOSFET can sustain damage from excessive power dissipation and heating. To limit any damage to the external MOSFETs or the motor, the gate-drive outputs should be fully disabled directly, either from the MCU or the PWM generator.

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Fault Action for GDU

The action taken when one of the diagnostic functions indicates a fault is listed in Table 6.

When a fault is detected, a corresponding fault state is considered to exist. In some cases, the fault state only exists during the time the fault is detected. In other cases, when the fault is only detected for a short duration, the fault state is latched (held in the fault state) until reset. The faults that are latched are indicated in Table 6. Any fault bits that have been set in the diagnostic registers remain until reset.

For most of the diagnostic functions, when a fault state is detected, the subsequent action can be programmed to force the gate-drive output to enter the inactive (low) state or to remain active. This action is selected by setting a 1 or a 0 in the specific stop-on-fault (SOF) bit associated with the diagnostic. The specific SOF bits for each diagnostic and the actions taken for each setting are listed in Table 6.

If a power-on-reset state is detected, all gate-drive outputs are driven low and all MOSFETs in the bridge are held in the off state. This persists until the power-on-reset state is cleared.

Setting any of the FVRU, FVBU, FDSO, FVSU, or FOT bits in the stop-on-fault registers to 0, such that the gate-drive outputs are not disabled in the event of the corresponding fault being detected, means that the GDU does not take any action to protect itself or the external bridge MOSFETs, and damage might occur. Appropriate action must be taken by the MCU.

FAULT MASKS

Individual diagnostics, except power-on reset, EEPROM error and over temperature, can be disabled by setting the corresponding bit in the mask register. Power-on-reset cannot be disabled because the diagnostics and the output control depend on correct operation of the logic regulator. If a bit is set to one in the mask register, the corresponding diagnostic are completely disabled. Fault states for the disabled diagnostic are not generated, and fault flags and diagnostic bits do not become set.

Care must be taken when diagnostics are disabled to avoid potentially damaging conditions.

Table 6: Fault Actions

Fault Description	SOF Bit Name	Disable Outputs		Fault State Latched
		SOF Bit = 0	SOF Bit = 1	
No Fault	—	No	—	—
Power-on-Reset	—	Yes [1]	No	—
EEPROM	—	Yes [1]	Yes [1]	Yes
VREG Undervoltage	FVRU	No [3]	Yes [1]	No
VLR Undervoltage	FVLU	Yes [1]	Yes [1]	No
Bootstrap Undervoltage	FVBU	Yes [2]	Yes [1]	Yes
ENABLE WD Timeout	FEN	No	Yes [1]	No
Overtemperature	FOT	No [3]	Yes [1]	No
Temperature Warning	FTW	No	Yes [1]	No
VDS Fault	FDSO	Yes [2]	Yes [1]	Yes
VBRG Undervoltage	FVSU	No [3]	Yes [1]	No

[1] All gate drives low, all MOSFETs off

[2] Gate drive for the affected MOSFET low, only the affected MOSFET off

[3] Stated fault condition might damage the A89201 and/or bridge MOSFETs

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MCU Diagnostics

Diagnostics in the MCU include external analog regulator monitors, such as the VD50 terminal undervoltage, overvoltage, VD33 terminal undervoltage, and VIO terminal undervoltage.

An internal clock-monitor feature is also included within the MCU. An internal independent oscillator monitors the external clock. If a fault occurs, the A89201 automatically selects the internal clock source to keep the program running.

The A89201 has a memory error detection and correction capability. The flash memory and RAM for both IRAM and DRAM are protected by an ECC algorithm. When a single error is detected, the contents are automatically corrected. When a double error is detected, it is reported; in some cases, performance of a reset might also be possible. For further details, refer to the [SCU user manual](#) available for download from the Allegro Software Portal. [1]

VD50 UNDERVOLTAGE

The VD50 terminal (the main supply voltage input to the MCU) has undervoltage detection. When the voltage reduces to less than the threshold level, $V_{VD50_UV_fal}$, the VD50 terminal undervoltage is detected and flagged to the system control unit.

Once the fault is removed and the voltage increases to greater than the threshold, $V_{VD50_UV_ris}$, the fault clears. The fault flag, SYSCON.VD50UV, remains until it is cleared.

VD50 UNDERVOLTAGE RECOVERY ($V_{VD50_OV_RIS}$)

The VD50 terminal, main supply voltage input to the MCU has VD50 voltage recovery detection. When the voltage increases to greater than the threshold level, $V_{VD50_OV_ris}$, the VD50 terminal voltage recovery is detected and flagged to the system control unit. The fault flag, SYSCON.VD50OV, remains until it is cleared.

VD33 TERMINAL UNDERVOLTAGE (V_{VD33_UV})

The VD33 terminal, analog internal supply voltage output to the MCU has undervoltage detection. When the voltage reduces to less than the threshold level, $V_{VD33_UV_fal}$, the VD33 terminal undervoltage is detected and flagged to the system control unit.

Once the fault is removed and the voltage increases to greater than the threshold, $V_{VD33_UV_ris}$, the fault clears. The fault flag, SYSCON.VD33UV, remains until it is cleared.

VIO TERMINAL UNDERVOLTAGE (V_{VIO_UV})

The VIO terminal (analog internal supply voltage output when 3.3 V IO voltage is selected; analog internal supply voltage input when 5 V IO voltage is selected) has undervoltage detection. When the voltage reduces to less than the threshold level, $V_{VIO_UV_fal}$, the VIO terminal undervoltage is detected and flagged to the system control unit.

After the fault is removed and the voltage increases to greater than the threshold, $V_{VIO_UV_ris}$, the fault clears. The fault flag, SYSCON.VIOUV, remains until it is cleared.

DATA DRAM ECC SINGLE ERROR

DRAM has error-correction code (ECC) capability. When ECC logic detects a single-bit error, a flag is reported in the system control unit, SYSCON.DSSEC. The ECC logic automatically corrects the faulty bit and function continues.

The flag remains until it is cleared.

DATA DRAM ECC DOUBLE ERROR

DRAM has ECC capability. When ECC logic detects a double-bit error, a flag is reported in the system control unit, SYSCON.DSDED. Once detected, if enabled, the system control unit interrupt can be called.

The flag remains until it is cleared.

INSTRUCTION IRAM ECC SINGLE ERROR

IRAM has ECC capability. When ECC logic detects a single-bit error, a flag is reported in the system control unit, SYSCON.ISSEC. The ECC logic automatically corrects the faulty bit and function continues.

The flag remains until it is cleared.

INSTRUCTION IRAM ECC DOUBLE ERROR

IRAM has ECC capability. When ECC logic detects a double-bit error, a flag is reported in the system control unit, SYSCON.ISDED. Once detected, if enabled, the system control unit interrupt can be called.

The flag remains until it is cleared.

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WATCHDOG TIMER OVERFLOW

The system control unit has an internal watchdog timer to monitor software activity. When a timeout is detected, if the system control unit interrupt is enabled, it is called.

For further details, refer to the [SCU user manual](#) available for download from the Allegro Software Portal. [1]

PLL LOSS OF LOCK

A89201 has an internal phase-locked loop (PLL) that generates the clock frequency, 80 MHz. The PLL block diagram is shown in Figure 25. The internal diagnostic continuously monitors the PLL. When the fault occurs and PLL ceases to be available to provide the intended clock frequency, the internal clock management automatically switches to the internal clock to maintain the operation; however, the clock frequency is 2.4 MHz. When this occurs, the fault flag is recorded in the system control unit, DIAG.PLLOK.

When the fault is removed, the clock automatically reverts to the PLL clock and the fault is removed. The fault flag remains until cleared.

When the SCU interrupt is enabled and INTEN.LOSTLOCK is enabled, interrupt is called to perform an appropriate action.

PLL TIMEOUT DETECTION

PLL timeout detection, which is a subsystem of the loss of PLL lock feature, enables the performance of more-detailed diagnostics. PLL timeout detection monitors the PLL lock status and provides a filter feature, which are used to determine if the duration of the PLL unlock status exceeds 1 ms. The PLL unlock state might be a temporary occurrence, and further diagnostics can be made by using PLL timeout detection.

When it detects that the duration of the PLL unlock status exceeds 1 ms, DIAG.PLLTO in the system control unit is flagged. The flag remains until it is cleared.

PLL SOURCE TIMEOUT DETECTION

PLL has an input clock monitor to ensure correct operation of the clock source to the PLL. The external clock is continually monitored using an independent internal oscillator.

When the PLL input frequency is out of range, DIAG.SRCTO in the system control unit is flagged. When DIAG.SRCTO is flagged, DIAG.PLLOK is reset.

If the input clock frequency enters the frequency range, the fault clears; however, the flag in the system controller remains until it is cleared.

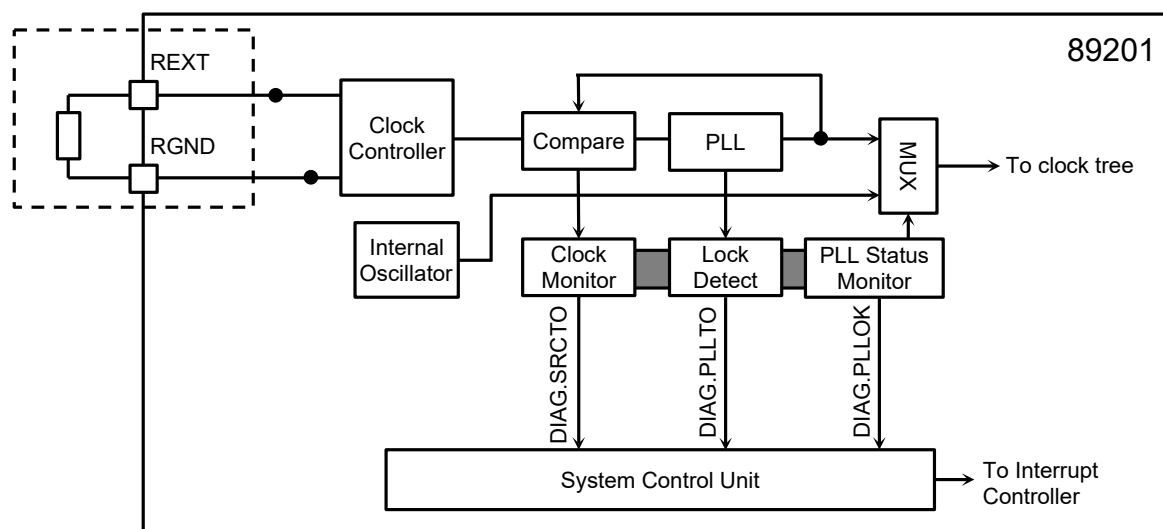
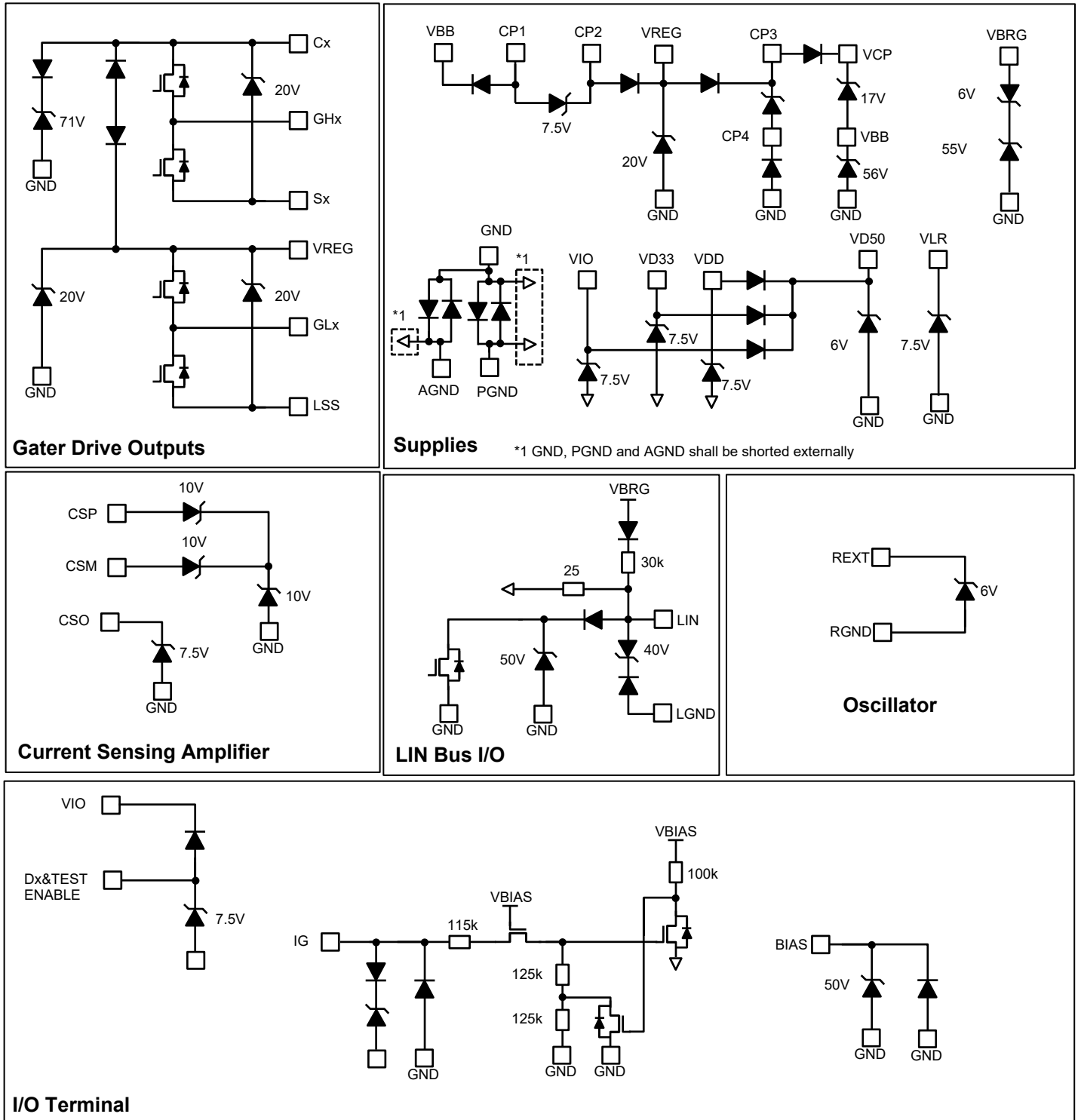


Figure 25: Block Diagram of PLL

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INPUT/OUTPUT STRUCTURES



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PACKAGE OUTLINE DRAWING

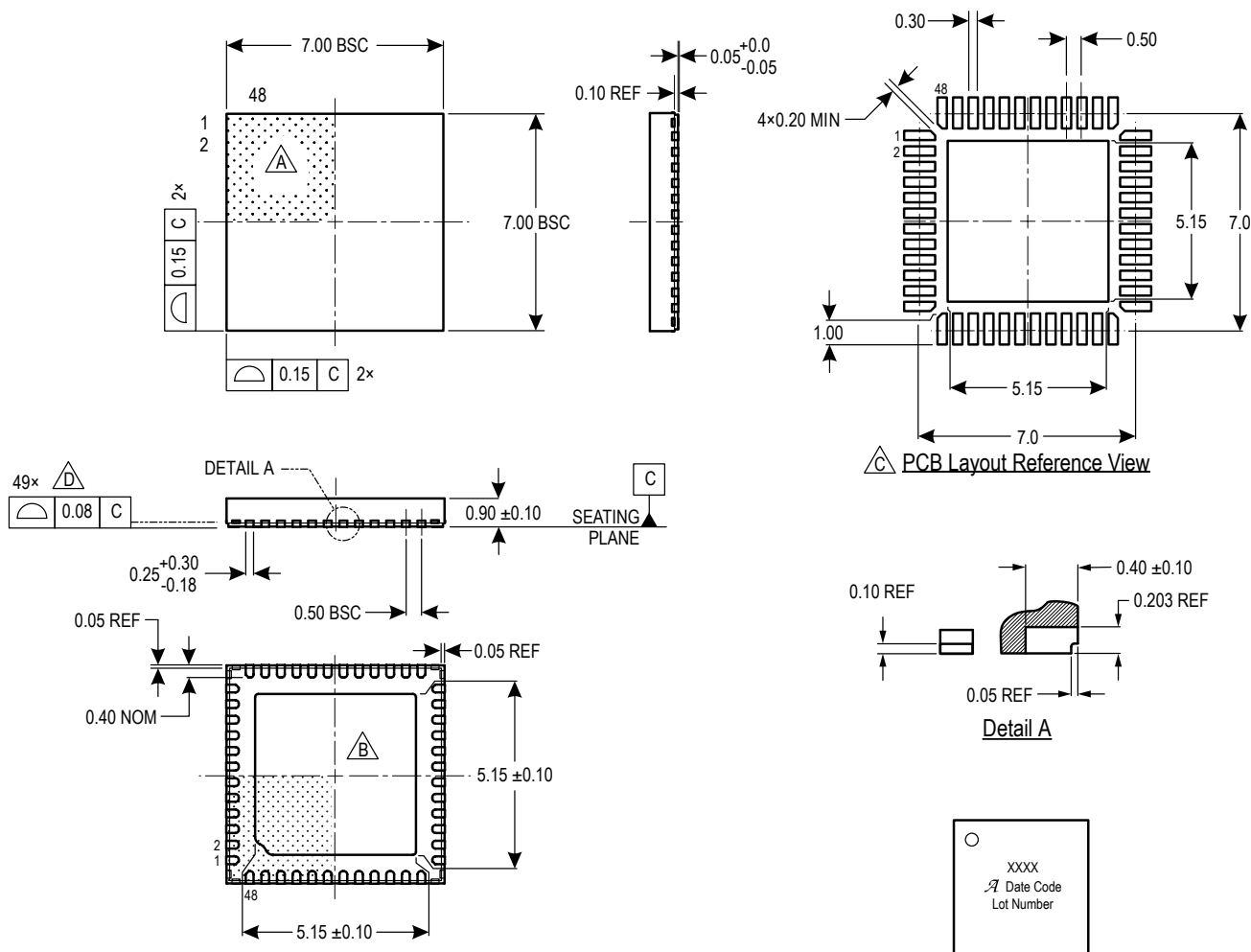
For Reference Only – Not for Tooling Use

(Reference DWG-0000378, Rev. 3)

Dimensions in millimeters

NOT TO SCALE

Exact case and lead configuration at supplier discretion within limits shown



- A** Terminal #1 mark area
- B** Exposed thermal pad (reference only, terminal #1 identifier appearance at supplier discretion)
- C** Reference land pattern layout (reference IPC7351 QFN50P700X700X100-49M); all pads a minimum of 0.20 mm from all adjacent pads; adjust as necessary to meet application process requirements and PCB layout tolerances; when mounting on a multilayer PCB, thermal vias at the exposed thermal pad land can improve thermal dissipation (reference EIA/JEDEC Standard JESD51-5)
- D** Coplanarity includes exposed thermal pad and terminals
- E** Branding scale and appearance at supplier discretion

- E** Standard Branding Reference View
 - Line 1: Part Number
 - Line 2: Logo A, 4-digit Date Code
 - Line 3: Characters 5, 6, 7, 8 of Assembly Lot Number

Figure 26: 48-Pin QFN with Exposed Pad (Suffix -EV)

A89201C**Automotive Microcontroller
with LIN and MOSFET Driver****Revision History**

Number	Date	Description
–	December 18, 2024	Initial release
1	January 2, 2025	Updated notes for pull-up peak source current and pull-down peak sink current characteristics (page 17)
2	February 26, 2025	Modified functional block diagram (page 1), selection guide footnote (page 4), and current sense input characteristic for maximum input bias current (page 9); added EEPROM and Flash characteristics (page 9); modified CSO current sense amplifier ground-referenced output characteristic for DC common-mode rejection (page 11), DAU characteristic for typical LSB (page 12), analog multiplexer inputs characteristics for regulator voltage (VCP) measure range and voltage (IG) measure range (page 13), and names of analog multiplexer inputs characteristics for AOUT divider ratios (page 13); added analog multiplexer inputs (AOUT) characteristic for additional AOUT divider ratio (VCP) (page 13); modified sensor bias terminal characteristic for minimum bias output voltage (page 14); modified names of gate output drive characteristics turn-on time step size and turn-off time step size; removed diagnostics and protection characteristic for VBB startup voltage (page 19); modified diagnostics and protection characteristics for minimum VDS qualify time (page 20); modified bus address map figure and RAM description (page 25); and added interactive pdf tags throughout
3	March 13, 2025	Minor editorial updates
4	July 21, 2025	Modified pedestal voltage drift characteristic of the CSO current sense amplifier of the ground-referenced output (page 11), turn-on rise time characteristic of the gate output drive (page 17), and CAU block diagram (page 29)
5	January 22, 2026	Updated selection guide table (page 4), Turn-Off Propagation Delay maximum value, Turn-On Propagation Delay maximum value, and Propagation Delay Matching (GHx-to-GLx) maximum value (page 18)

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